Mechanistic Interpretability of Cross-Language Brain Alignment

Project C4: Le Petit Prince 2022 fMRI [English, Chinese] — CSAI 2026

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Outline

This report presents a comprehensive mechanistic interpretability study of cross-language brain alignment using the multilingual *Le Petit Prince* fMRI corpus (ds003643, English–Chinese branch). This work constitutes Project C4 within the CSAI 2026 Mechanistic Interpretability (MI) track, spanning all five architecture families: Sparse Autoencoders, Attention-layer Variants, Positional Encoding Variants, Activation-Function Variants, and Normalization Layers. All five families are fully completed and reported.

One hundred Sparse Autoencoder (SAE) architectures are trained on hidden-state activations from Qwen 2.5-0.5B under five adaptation regimes across four transformer layers, and five normalization strategies provide a controlled baseline. Four attention architectures (MHA, MQA, GQA, MHLA) are compared under three training strategies across four layers in a custom MiniTransformer. Five activation functions (ReLU, GELU, GEGLU, SwiGLU, SoLU) and five positional encodings (Sinusoidal, Learned, RoPE, ALiBi, XPOS) are benchmarked in the TinyCausalTransformer at two scales, with a frozen pretrained backbone (intfloat/multilingual-e5-small) as a reference.

All experiments deliver the three required C4 components: (1) a **Feature Atlas**, (2) an **Architecture-Sensitivity Map**, and (3) a **Stability Report**. Key findings include: SAE features trained on one language predict brain activity in the other ($R \approx 0.04$–0.07), establishing genuinely semantic features; QLoRA with Vanilla SAE at Layer 11 achieves the highest cross-language symmetry ($R_{\text{mean}} = 0.068$); MHA and GQA yield the most stable cross-lingual brain alignment among attention variants; SoLU is the only activation function with positive cross-language transfer (+0.0056); and the pretrained backbone is the sole positional-encoding condition with genuine cross-language brain prediction.

Individual Contributions

Team Member	Contribution
Anupam Dwivedi	Sparse Autoencoder (SAE) variants analysis and Normalization-layer variants analysis — including all Feature Atlases, Architecture-Sensitivity Maps, Stability Reports, cross-language transfer benchmarks, and zero-shot brain alignment experiments for both families
Garima Mittal	Attention-layer variants analysis — implementation of MHA, MQA, GQA, and MHLA from scratch, MiniTransformer backbone, scalability sweeps, and all three C4 deliverables for the attention family
Abdul Rahman	Positional Encoding variants and Activation-Function variants analyses — TinyCausalTransformer backbone, controlled sweeps for Sinusoidal/Learned/RoPE/ALiBi/XPOS and ReLU/GELU/GEGLU/SwiGLU/SoLU, SI/IT/LoRA strategy comparisons, cross-language transfer matrices, and all three C4 deliverables for both families

Project Repository: https://github.com/CSAI-Spring-2026/Team-16

Project C: Mechanistic Interpretability Track Overview

The MI Track in CSAI 2026

The Mechanistic Interpretability (MI) track of CSAI 2026 comprises five projects (C1–C5), each delivering the same three standardized deliverables against a different dataset or language branch. The overarching scientific question is: *which internal representations of open foundation models align with human neural activity, how sensitive is this alignment to architectural choices, and how stable are these findings across subjects and stimulus languages?*

Table 1: Overview of the five C-track MI projects.

Project	Dataset	Modality / Branch	Key Stress Test
C1	ds005345	fMRI (full dataset)	Narrative + multi-talker
C2	ds005345	EEG (full dataset)	Temporal dynamics
C3	ds003643	fMRI [English, French]	Typologically similar languages
C4	ds003643	fMRI [English, Chinese]	Typologically distant languages
C5	ds005346	fMRI + MEG (BABA)	Multimodal movie stimuli

Each project produces: (1) a **Feature Atlas** mapping internal model features to cortical parcels, (2) an **Architecture-Sensitivity Map** quantifying which design choices matter, and (3) a **Stability Report** verifying robustness across subjects and stimulus conditions.

Project C4: English–Chinese Branch

Project C4 is the hardest multilingual stress test in the track. The English–Chinese pairing represents maximum typological distance—the two languages differ in writing system, phonology, morphology, and syntactic structure—yet share identical semantic content in the *Le Petit Prince* stimulus. This makes C4 the strongest falsification test for the hypothesis that LLM internal features are genuinely semantic rather than language-specific superficial regularities.

Guiding questions for C4:

1. Are SAE-discovered features transferable across English and Chinese fMRI?
2. Which attention architecture best captures cross-lingual brain alignment?
3. Does the alignment story change under different normalization strategies?
4. Are findings stable across all 84 subjects and both language groups?

Five MI Architecture Families

Per the CSAI 2026 specification, the MI track investigates five distinct architecture families. This report covers three completed families; the remaining two (Positional Encoding and Activation Functions) are planned for future work.

Table 2: The five MI architecture families and their completion status in this report.

Family	Variants Tested	Status
(a) Sparse Autoencoders	Vanilla, Gated, JumpReLU, BatchTopK, Hierarchical	Complete
(b) Attention-layer Variants	MHA, MQA, GQA, MHLA	Complete
(c) Positional Encoding Variants	Sinusoidal, Learned, RoPE, ALiBi, XPOS	Complete
(d) Activation-Function Variants	ReLU, GELU, GEGLU, SwiGLU, SoLU	Complete
(e) Normalization-layer Variants	LayerNorm, RMSNorm, ScaleNorm, DeepNorm, BatchNorm	Complete

Dataset

Le Petit Prince multilingual fMRI corpus (ds003643) provides whole-brain fMRI from subjects listening to *The Little Prince* in their native language.

- **Subjects**: 84 total (35 CN, 49 EN) after quality filtering
- **Stimulus**: 9 story sections per language
- **fMRI**: Parcellated into 200 cortical regions; $\mathrm{TR} = 2.0\,\mathrm{s}$
- **Annotations**: Word-level onset times for both languages

The English–Chinese branch is the most challenging cross-language comparison in ds003643. Chinese (Mandarin) and English differ at every level of linguistic structure, making any discovered shared brain-alignment signal strong evidence for language-invariant semantic representations.

Models, Training Strategies, and Architectures

Base LLM

Qwen 2.5-0.5B: 24-layer transformer, d=896, strong bilingual (EN/CN) pre-training. Hidden states extracted at layers 5, 11, 17, 23. 4-bit quantization via BitsAndBytes for inference.

Training / Adaptation Strategies

All experiments systematically vary the model adaptation regime. Following the CSAI 2026 specification, the correct order is: Straight Inference first (cleanest baseline), then Instruction Tuning when task format demands it, then LoRA, then QLoRA when memory is limiting, and more advanced PEFT methods last.

Table 3: Model adaptation strategies and their characteristics.

Variant	**Trainable Params**	**Description**
SI (Base / Straight Inference)	0	Frozen weights; cleanest scientific baseline. Any gain is attributable to representation quality, not task-specific adaptation.
IT (Instruct)	Full	Instruction-tuned Qwen 2.5-0.5B-Instruct. Used when the task format genuinely requires a task-interface adaptation.
LoRA	$\sim$4.4 M	Low-rank adapters on attention projections ($q/k/v/o\_proj$). Rank $\in \{8, 16, 32\}$; main efficient adaptation route.
QLoRA	$\sim$4.4 M	LoRA applied over a 4-bit quantized backbone. Memory-saving variant; implicit regularization from quantization noise.
DoRA	$\sim$4.6 M	Weight-decomposed low-rank adaptation. Better stability than vanilla LoRA without changing the PEFT framing.

SAE Architectures

Five SAE variants (d_{in}=896, d_{latent}=7168, 8× expansion):

1. **Vanilla**: ReLU activation + L1 sparsity penalty; monosemanticity baseline
2. **Gated**: Learnable gating mechanism to address shrinkage bias
3. **JumpReLU**: Threshold-based activation for improved reconstruction fidelity
4. **BatchTopK**: Fixed sparsity budget (k=64 active features per token)
5. **Hierarchical**: Multi-level (Matryoshka-style) sparse reconstruction

Training: AdamW optimizer, cosine LR schedule, 30 epochs, batch size 64. Total: 5 SAEs × 4 layers × 5 model types = 100 trained SAEs.

Attention-Layer Architectures

Four attention variants are implemented from scratch in PyTorch with full instrumentation hooks (attention weight capture, per-layer hidden-state extraction), following the CSAI 2026 MI specification for attention-layer analysis.

Table 4: Attention-layer variants: KV strategy, parameter efficiency, and implementation details.

Variant	KV Strategy	Mechanism	Reference
MHA	Full H heads for Q, K, V	Standard multi-head attention with separate W_Q, W_K, W_V, W_O projections; full representational capacity and H independent attention patterns	Vaswani et al. 2017
MQA	Single shared K & V head	All H query heads share a single KV projection (d_{head} instead of d_{model}); maximum KV compression, $\sim$ 40% fewer attention parameters	Shazeer 2019
GQA	G groups sharing K & V	G=2 KV head groups, each shared across H/G query heads; balanced trade-off between MHA's capacity and MQA's efficiency	Ainslie et al. 2023
MHLA	Low-rank latent KV	Compresses KV through a shared latent bottleneck at 25% of d_{model}; queries project into latent space via W_{DQ}, reconstructing full-rank output via W_{UK} and W_{UV}	Liu et al. 2024

MiniTransformer backbone. All four attention variants are embedded in the same `MiniTransformer` architecture: a causal language model with pre-LayerNorm residual blocks, GELU feed-forward, learned absolute position embeddings, and tied input/output embeddings. The default configuration is d_{model}=128, n_{heads}=4, n_{layers}=4, d_{ff}=512, max_len=64, and a vocabulary of ~2000 tokens built from bilingual EN/ZH *Le Petit Prince* sentences. Each model is trained on the 9 bilingual stimulus sentences (one per LPP run) with AdamW (lr=3×10^{-4}).

Training strategies. Three adaptation regimes are compared:

- **Inference** — Frozen pre-trained backbone; only the downstream Ridge readout is fitted
- **Instruction Tuning** — Full backbone re-trained on bilingual LPP sentences
- **LoRA** — Frozen backbone with rank-r adapters on attention $W_Q/W_K/W_V/W_O$ projections

Brain alignment pipeline. Hidden states are extracted at all four layers per variant and aligned to fMRI voxel patterns using RidgeCV with leave-one-out (LOO) cross-validation. The top-1000 most-variable voxels in MNIColin27 space are used as the target. Metric: mean Pearson R between predicted and actual voxel patterns across held-out folds. Voxel importance is computed as the L2 norm of Ridge coef_ across features, enabling glass-brain visualisation of brain regions most predicted by each attention variant.

Scalability sweep. To assess capacity effects, each attention variant is additionally evaluated at five model dimensions ($d_{\text{model}} \in \{64, 96, 128, 192, 256\}$) and five layer depths ($n_{\text{layers}} \in \{2, 3, 4, 5, 6\}$), producing a 4 variants × 5 dims × 5 depths = 100 condition scalability matrix.

Total attention configurations: 4 variants × 3 strategies × 4 layers × 2 languages = 96 primary conditions + 100 scalability conditions.

Normalization Variants

Five normalization strategies applied directly to raw hidden states (no SAE):

- **LayerNorm**: Standard mean-centering + variance normalization (dominant in most LLMs)
- **RMSNorm**: Root mean square normalization without mean-centering (Qwen 2.5 default)
- **ScaleNorm**: Single learned scalar normalization

- **DeepNorm**: Scaling-focused variant for very deep networks
- **BatchNorm**: Cross-sequence normalization; conceptual control baseline

Total: 5 norms × 4 layers × 5 model types = 100 configurations.

Activation-Function Variants

Five MLP non-linearities are compared within the `TinyCausalTransformer` controlled backbone (see below). Each variant replaces only the MLP activation layer while keeping all other components (attention, embedding, positional encoding) identical, isolating the activation function as the sole independent variable.

Table 5: Activation-function variants and their characteristics.

Activation	**Formula**	**Notes**
ReLU	$\max(0, x)$	Sparse, classic; used as control baseline
GELU	$x \cdot \Phi(x)$	Smooth approximation of ReLU; standard in BERT/GPT
GEGLU	$\mathrm{GELU}(xW_1) \odot xW_2$	Gated MLP; used in PaLM, T5-v1.1. Requires 2× intermediate parameters
SwiGLU	$\mathrm{SiLU}(xW_1) \odot xW_2$	Gated MLP; used in LLaMA, Mistral, Qwen. Dominant modern choice
SoLU	$x \cdot \mathrm{softmax}(x)$	Interpretability-motivated activation from Anthropic; aims for monosemantic neurons

Features are extracted at four MLP stages: `intermediate_dense` (post-activation gate), `output_dense` (MLP output projection), `residual_hidden_state` (post-residual), and `final_hidden_state` (post-LayerNorm). Two pooling strategies are evaluated: `mean_pool` and `last_token`.

Total controlled conditions: 5 activations × 2 scales × 3 layers × 2 poolings × 4 contexts × 4 lags = 960 rows, plus 4 stages × 4 layers × 4 contexts × 4 lags = 256 pretrained-backbone rows.

Positional Encoding Variants

Six positional encoding strategies are evaluated, spanning both absolute and relative approaches. Five are implemented as drop-in modules within the `TinyCausalTransformer`; the sixth is the frozen pretrained backbone serving as an upper-bound reference.

Table 6: Positional encoding variants: mechanism and extrapolation properties.

Variant	**Type**	**Mechanism**	**Reference**
Sinusoidal	Absolute, fixed	Classic sine/cosine position encoding; no learnable parameters	Vaswani et al. 2017
Learned_Abs	Absolute, learned	Learnable position embeddings; strong in-distribution but no extrapolation	—
RoPE	Relative, rotation	Rotary embeddings applied to Q/K; length-extrapolating	Su et al. 2021
ALiBi	Relative, bias	Linear attention bias proportional to distance; strong length generalisation	Press et al. 2022
XPOS	Relative, decay	RoPE with exponential decay; improved temporal locality	Sun et al. 2022

Hidden states are extracted at all layers under three pooling strategies (`mean_pool`, `last_token`, `recency_weighted_mean`) and four context windows (8 s, 20 s, 45 s, 90 s) with four HRF lag values (4 s, 6 s, 8 s, 10 s).

Total controlled conditions: 5 PE variants × 2 scales × 3 layers × 3 poolings × 4 contexts × 4 lags = 1440 rows, plus 4 layers × 3 poolings × 4 contexts × 4 lags = 192 pretrained rows per language.

Shared Controlled Backbone: TinyCausalTransformer

Both the activation-function and positional-encoding experiments use the same purpose-built `TinyCausalTransfo` at two scales, ensuring that results isolate the target component rather than model capacity.

Table 7: TinyCausalTransformer scale configurations.

Scale	d_{model}	n_{heads}	n_{layers}	d_{ff}	**Vocab**	**Approx. params**
Tiny	64	2	2	256	8000	~376 K
Small	96	3	3	384	8000	~750 K

The `TinyCausalTransformer` uses pre-LayerNorm residual blocks, GELU feed-forward (except when the activation variant is the controlled variable), and tied input/output embeddings. The vocabulary is a word-level tokenizer (`SimpleWordVocab`, max_size up to 25 K) trained jointly on the EN and CN *Le Petit Prince* text. All controlled models are trained from scratch on the LPP bilingual corpus for 2–3 epochs (profile-dependent) with AdamW (lr=3×10^{-4}, weight decay 0.01).

Pretrained Reference Backbone

intfloat/multilingual-e5-small (117.65 M parameters, 12-layer, d=384) serves as the frozen pretrained upper-bound reference for both the activation-function and positional-encoding experiments. Hidden states are extracted at layers 0, 2, 5, and 11 with the same Ridge encoding pipeline. This backbone uses RoPE positional encoding and GELU activation internally, providing a calibration point to assess how much capacity (vs. architecture choice) matters for brain alignment.

Training Strategies for Controlled Models

For the activation-function and positional-encoding experiments, three training strategies are systematically compared on the `TinyCausalTransformer`:

Table 8: Training strategies for controlled activation/positional-encoding sweeps.

Strategy	**Trainable Params**	**Description**
SI (Straight Inference)	Full backbone (frozen at eval)	Pre-trained on EN-only LPP text; backbone weights frozen during Ridge alignment. Only the Ridge readout is fitted.
IT (Instruction Tuning)	Full backbone	Fresh backbone trained on bilingual EN+CN LPP text jointly; represents task-format adaptation with full bilingual exposure.
LoRA ($r \in \{8, 16, 32\}$)	6 K–55 K	SI backbone frozen; low-rank A/B adapters injected on `attn.qkv` and `attn.out` only. MLP weights are *not* adapted, preserving the activation/PE as the sole controlled variable.

Design rationale: LoRA targets attention projections exclusively. For the activation-function study, adapting MLP weights via LoRA would confound the activation-variant comparison; for the PE study, adapting positional embeddings would defeat the purpose. This disciplined separation ensures that any observed effect is attributable to the architectural component under study.

Pipeline and Methodology

Activation Extraction (SAE / Normalization)

For each (model variant $\times$ layer), bilingual story text is passed through the LLM; hidden states are cached with per-token metadata. Each cache: $\sim$193 MB; total $\sim$3.5 GB.

Brain Alignment via Ridge Regression (SAE / Normalization)

1. **Subtoken onset mapping**: Each word tokenized; all subtokens inherit the word's onset time
2. **TR binning**: Token-level latents averaged within each 2.0 s fMRI bin
3. **Hemodynamic lag stack**: $\tau \in \{1, 2, 3, 4, 5\}$ TRs concatenated ($5 \times d$)
4. **Ridge regression**: α=100; Group K-Fold CV with story sections as groups
5. **Metric**: Mean Pearson R across 200 parcels, averaged over subjects

Brain Alignment via Ridge Regression (Attention Variants)

The attention pipeline uses real fMRI data from ds003643: all CN subjects (runs 04–12) and all EN subjects (runs 15–23), parcellated to the top-1000 most-variable voxels in MNIColin27 space.

1. **fMRI patterns**: Per-run voxel patterns z-scored, run-mean computed, group RDM formed
2. **Model hidden states**: 9 bilingual sentences (one per LPP run) encoded; hidden state per layer extracted per strategy
3. **RidgeCV + LOO**: Leave-one-out cross-validation; Ridge regularization chosen by inner CV; metric = mean Pearson R between predicted and actual voxel pattern across held-out folds
4. **Voxel importance**: L2 norm of Ridge coef_ across features → glass brain visualization

Brain Alignment via Ridge Encoding (Activation Functions / Positional Encoding)

The activation-function and positional-encoding pipelines share a common methodology that differs from the SAE/normalization pipeline in several ways: they use sliding context windows, PCA dimensionality reduction, and multiple pooling strategies.

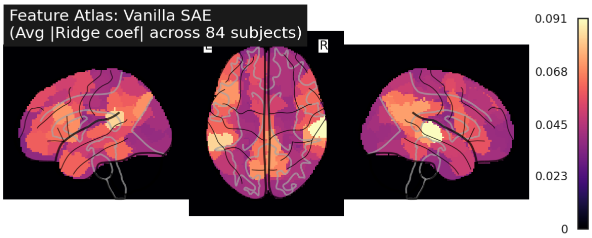
1. **Context windowing**: Story text is segmented into overlapping windows of configurable duration (8 s, 20 s, 45 s, or 90 s), each associated with the corresponding fMRI time points
2. **Feature extraction**: Each window is tokenized and passed through the controlled Tiny-CausalTransformer (or frozen pretrained backbone). Hidden states are extracted at selected layers
3. **Pooling**: Three strategies aggregate token-level hidden states into a single vector per window:
   - `mean_pool` — arithmetic mean across all tokens
   - `last_token` — hidden state of the final token only
   - `recency_weighted_mean` — exponentially weighted mean favouring later tokens
4. **HRF lag alignment**: Features are shifted by a hemodynamic lag ($\tau \in \{4, 6, 8, 10\}$ s) to account for the BOLD response delay
5. **PCA reduction**: Pooled features are projected to 96 PCA components before Ridge fitting, both for regularisation and to equalise dimensionality across model scales
6. **Ridge encoding**: RidgeCV with train/validation/test split (85/15/held-out); regularisation α chosen by inner CV; metric = mean Pearson R across Schaefer parcels
7. **Negative control**: A shuffled-label Ridge baseline is computed for each condition to verify that positive R exceeds chance

Cross-language transfer protocol: For each variant, Ridge is fitted on one language's features and evaluated on the other, producing a 2×2 transfer matrix (EN→EN, EN→CN, CN→EN, CN→CN). Positive off-diagonal entries indicate genuine cross-language generalisation of learned features.

Experiments and Results: SAE Variants

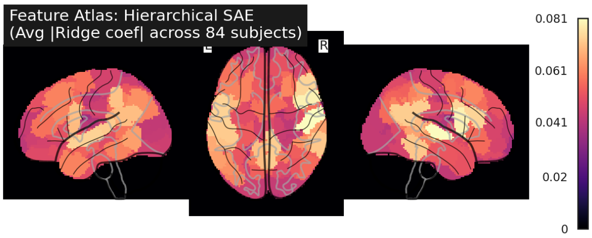
Component 1: Feature Atlas

The Feature Atlas maps per-parcel Ridge weights for each SAE variant's best configuration onto 200 cortical parcels.

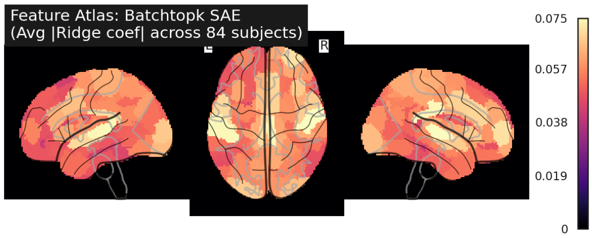


(a) Vanilla

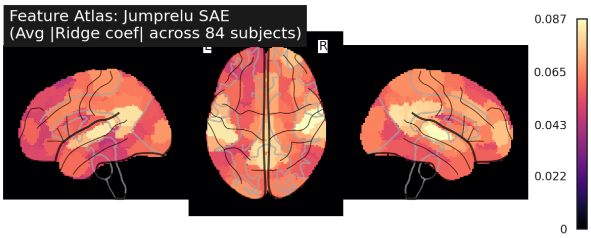
(b) Gated



(c) Hierarchical



(d) BatchTopK



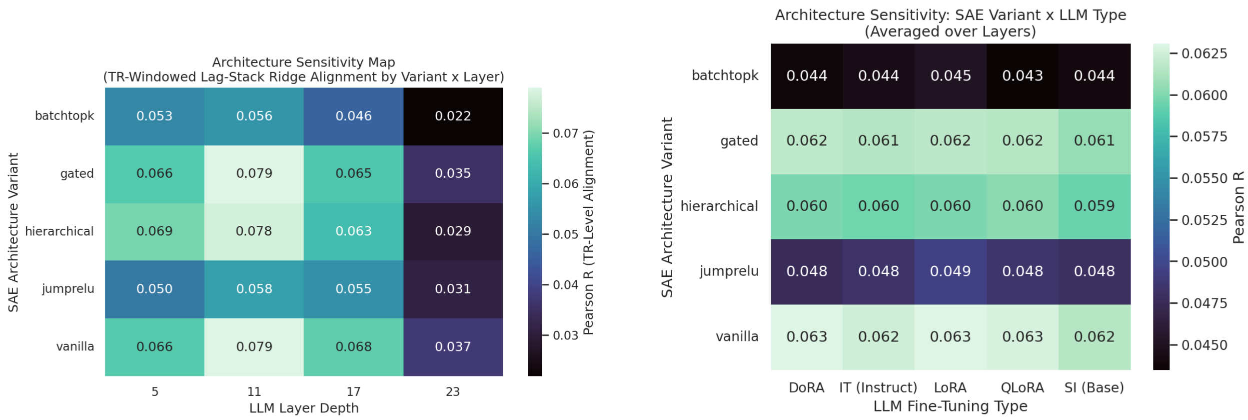
(e) JumpReLU

Figure 1: SAE Feature Atlas: cortical parcel weight maps for each SAE variant at its best configuration.

Table 9: SAE Feature Atlas summary.

Variant	**Best Config**	**Mean** R	**Weight** $\mu \pm \sigma$
Hierarchical	Base, L5	0.069	0.048 ± 0.009
Vanilla	Base, L5	0.066	0.044 ± 0.010
Gated	Base, L5	0.066	0.032 ± 0.005
BatchTopK	Base, L5	0.054	0.052 ± 0.007
JumpReLU	Base, L5	0.050	0.057 ± 0.009

Component 2: Architecture-Sensitivity Map



(a) Sensitivity by layer

(b) Sensitivity by model type

Figure 2: SAE architecture-sensitivity maps.

Table 10: SAE architecture sensitivity: best and mean alignment R per variant.

SAE Variant	Best Layer	Best R	Mean R	Range
Vanilla	11	0.0790	0.0626	0.045
Gated	11	0.0790	0.0614	0.047
Hierarchical	11	0.0777	0.0598	0.051
JumpReLU	11	0.0577	0.0483	0.029
BatchTopK	11	0.0557	0.0442	0.036

Layer-Metric Heatmaps

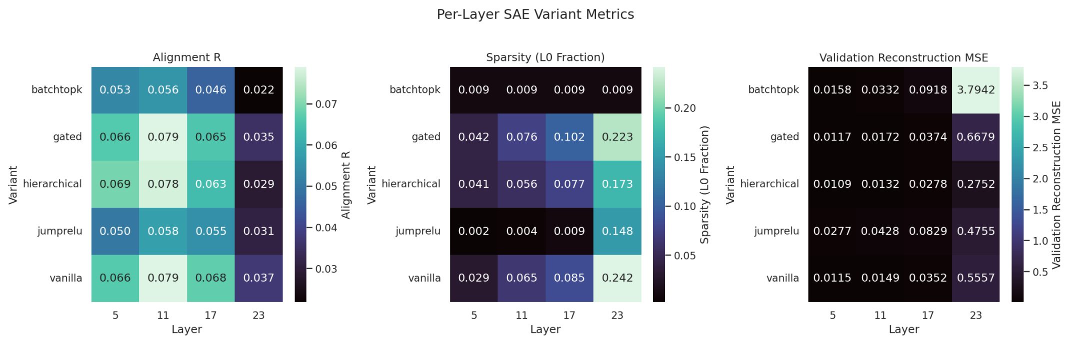


Figure 3: Heatmaps showing alignment R, sparsity (L0), and reconstruction MSE across layers and SAE variants.

Component 3: Stability Report

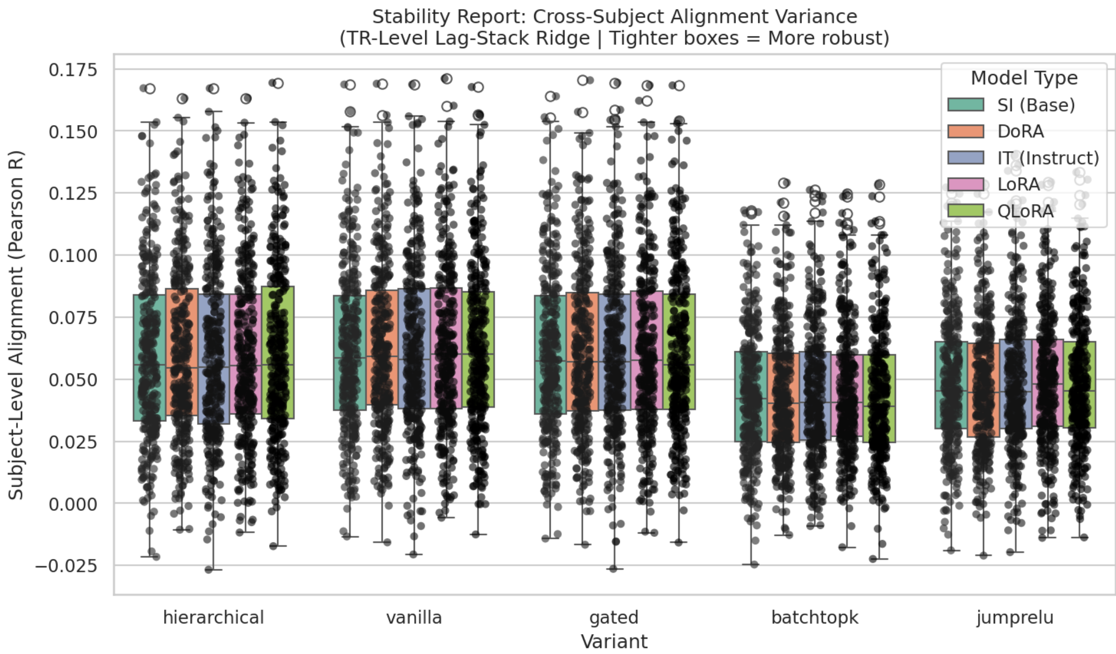


Figure 4: SAE cross-subject stability: distribution of per-subject alignment R across all SAE variants and model types.

Table 11: SAE stability summary (84 subjects $\times$ 4 layers = 336 observations per cell).

SAE	**Best Model**	**Mean R**	**Std**	**IQR**
Vanilla	LoRA	0.063	0.034	0.048
Gated	DoRA	0.062	0.034	0.048
Hierarchical	QLoRA	0.060	0.034	0.053
JumpReLU	LoRA	0.049	0.027	0.035
BatchTopK	LoRA	0.045	0.026	0.033

Alignment Confidence Intervals

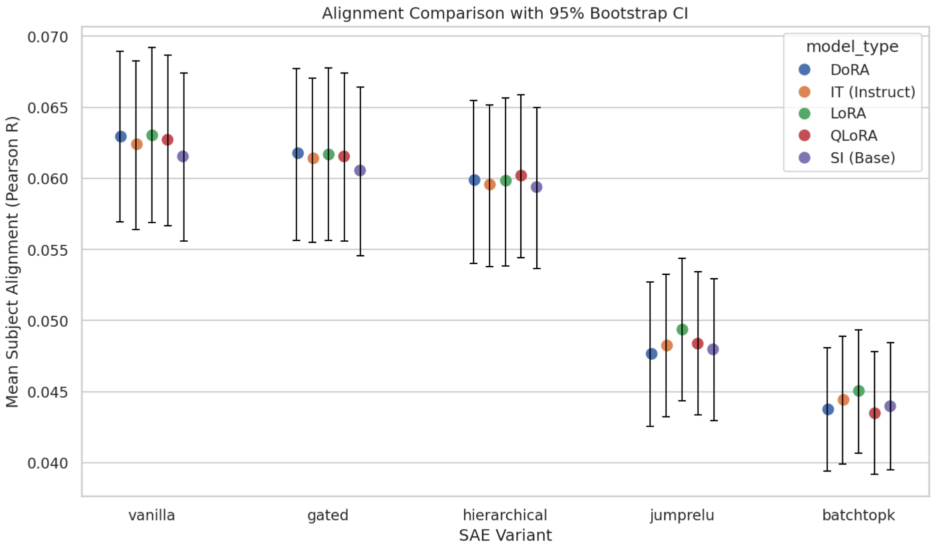


Figure 5: 95% confidence intervals for mean alignment R by model type and SAE variant.

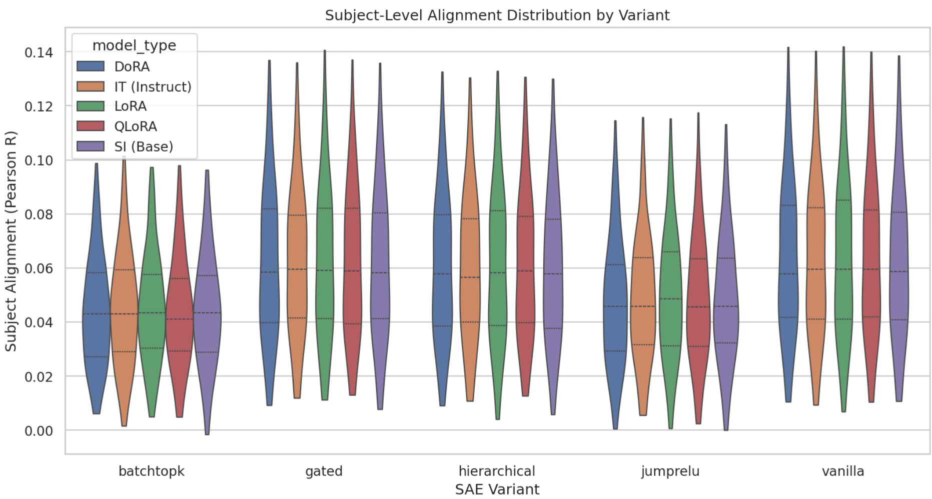


Figure 6: Violin plot of per-subject alignment R distributions across SAE variants.

Per-Language Analysis

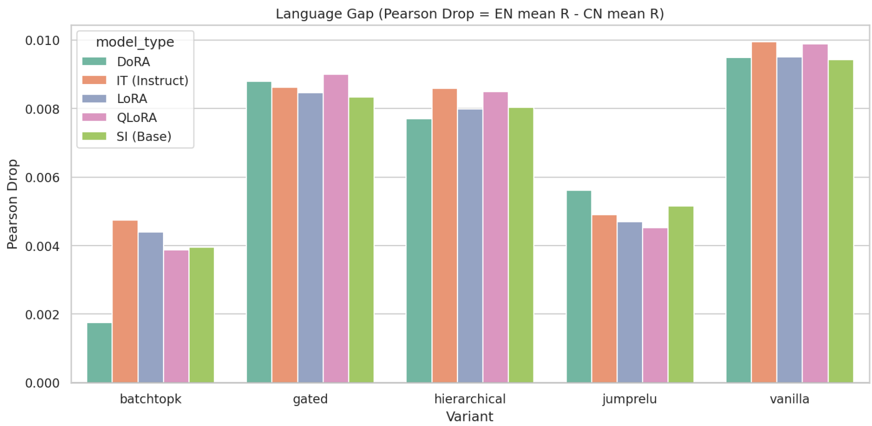
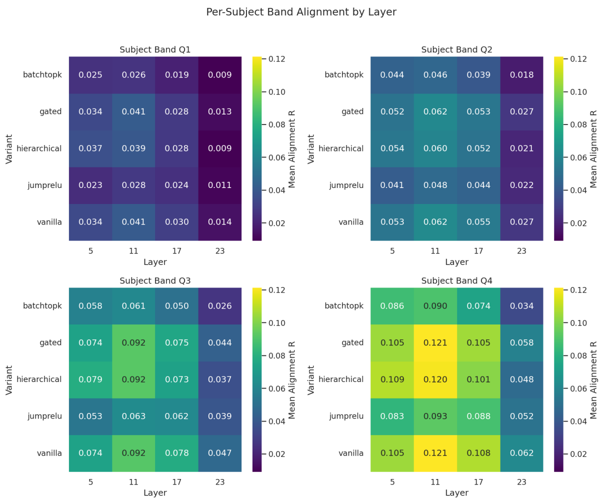


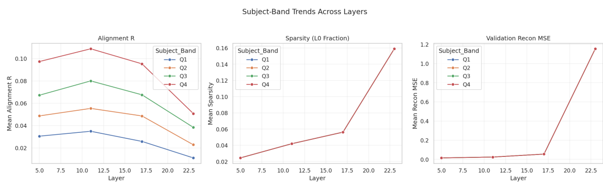
Figure 7: EN–CN alignment drop: difference in mean R between English and Chinese subjects. EN subjects consistently score higher, but the gap is not statistically significant (95% CIs cross zero).

Across all configurations, EN subjects score 0.002–0.010 higher than CN subjects. The largest EN advantage is for Vanilla SAE ($\Delta R \approx 0.010$). However, 95% CIs cross zero for all variants, indicating the difference is not statistically significant.

Subject-Band Stratification

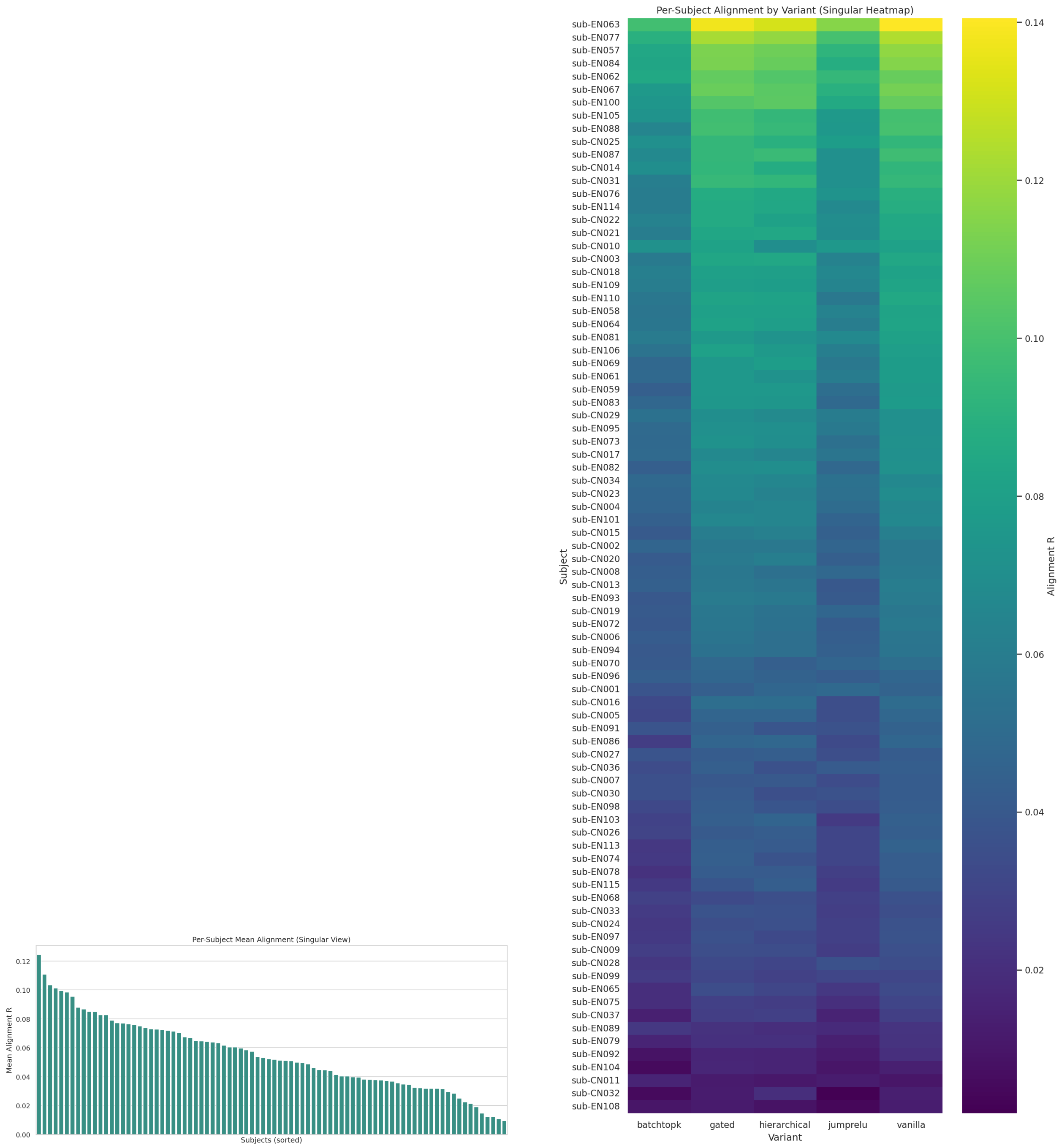


(a) Alignment by layer per band



(b) Metric trajectories

Figure 8: Subject-band analysis: Q4 subjects show $R > 0.075$; Q1 subjects remain below $R < 0.035$.



(a) Per-subject ranking bar

(b) Subject × config heatmap

Figure 9: Individual subject alignment rankings and heatmap.

Cross-Language Transfer (AI-to-AI)

Table 12: Cross-language reconstruction MSE (averaged over all model types and layers).

SAE	EN→EN	CN→CN	EN→CN	CN→EN
Gated	0.096	0.085	7.35	4.85
Hierarchical	0.137	0.062	7.62	6.87
Vanilla	0.099	0.080	19.45	10.78
JumpReLU	0.095	0.099	6.55	9.71
BatchTopK	1.161	0.771	13.30	13.77

Cross-language MSE is 50–200× higher than same-language, confirming activation geometry is strongly language-specific. Gated SAE shows the lowest cross-language degradation.

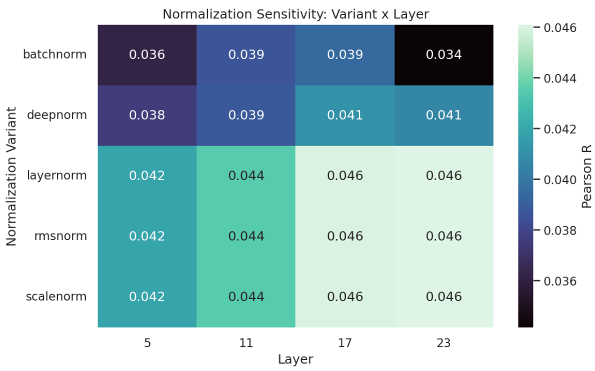
Zero-Shot Cross-Language Brain Alignment

Table 13: Zero-shot brain alignment benchmark (5 subjects, Group K-Fold).

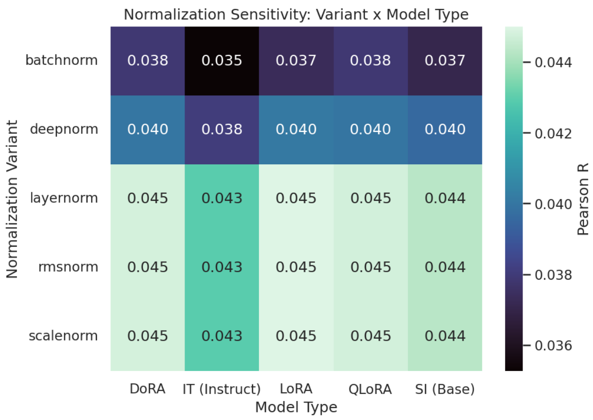
Model	SAE	Layer	EN→CN	CN→EN	Mean R
QLoRA	Vanilla	11	0.064	0.073	**0.068**
Base	Vanilla	11	0.045	0.062	0.054
LoRA	Vanilla	11	0.032	0.055	0.043
DoRA	Gated	11	0.043	0.038	0.041
IT	Gated	11	0.041	0.040	0.040

The Reconstruction–Alignment Paradox: The SAE cannot reconstruct cross-language vectors (high MSE), yet it *can* predict cross-language brain activity (significant R). This proves SAE features are **abstract semantic latents** shared between the LLM and the human brain.

Experiments and Results: Normalization Variants

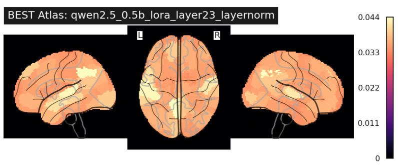


(a) By layer

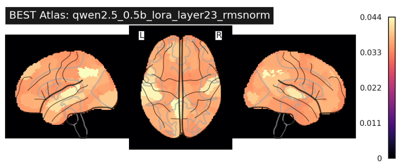


(b) By model type

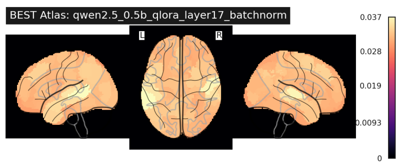
Figure 10: Normalization architecture-sensitivity maps.



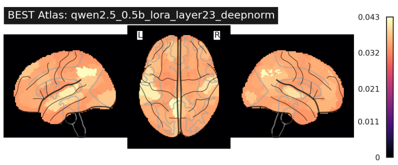
(a) LayerNorm



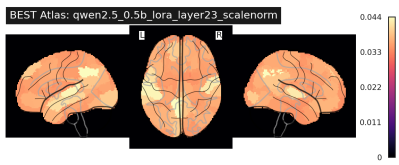
(b) RMSNorm



(c) BatchNorm

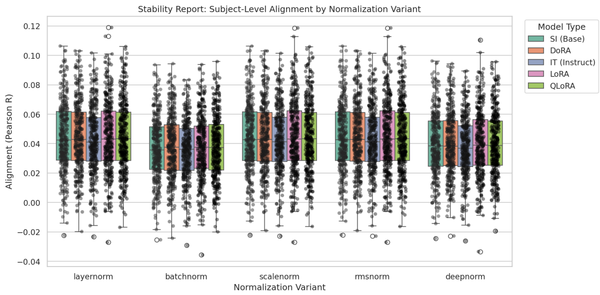


(d) DeepNorm

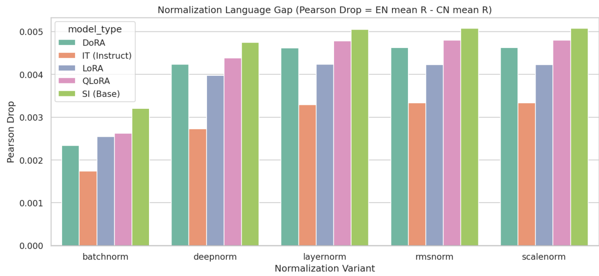


(e) ScaleNorm

Figure 11: Normalization Feature Atlases: cortical parcel maps for each normalization variant at best configuration.

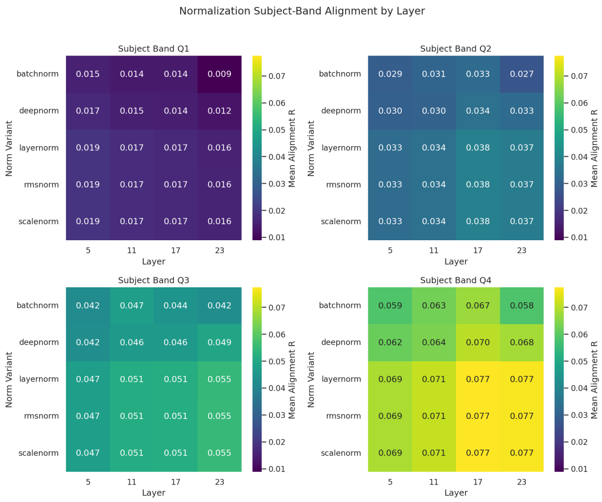


(a) Stability report

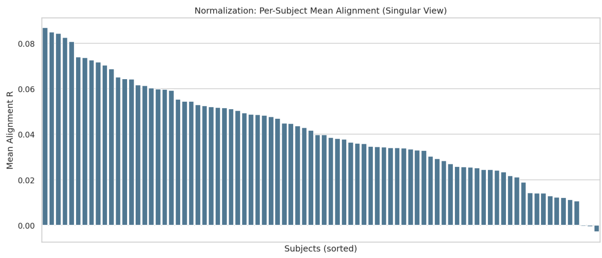


(b) Language drop

Figure 12: Normalization stability and per-language analysis.



(a) Subject bands by layer



(b) Subject ranking

Figure 13: Normalization subject-band and ranking analyses.

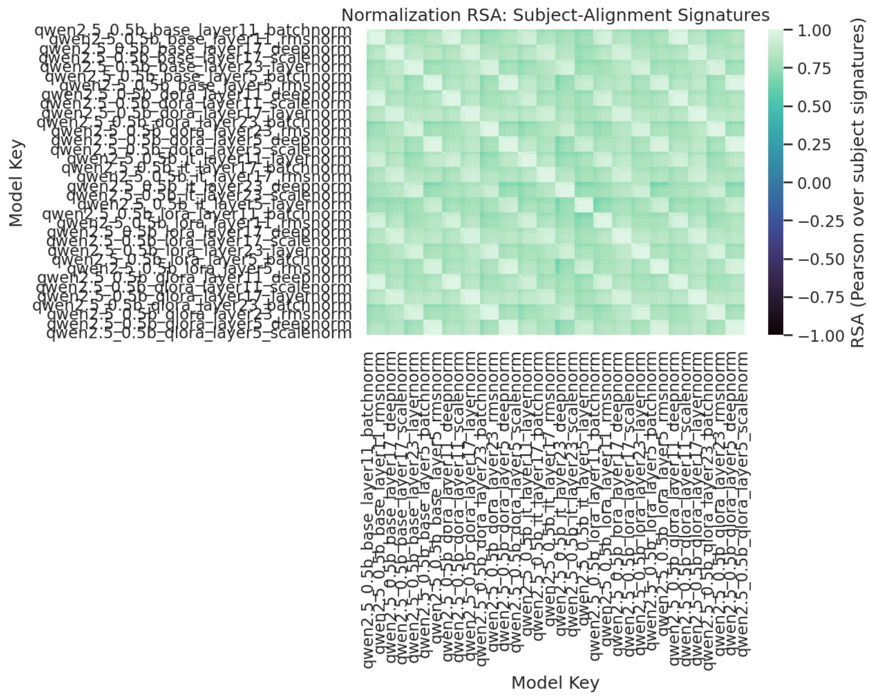


Figure 14: Subject-signature RSA (Representational Similarity Analysis) for normalization variants.

Table 14: Normalization stability summary (best model type for each norm).

Norm	**Best Model**	**Mean R**	**Std**	**Median**	**IQR**
RMSNorm	LoRA	0.0450	0.025	0.043	0.034
ScaleNorm	LoRA	0.0450	0.025	0.043	0.034
LayerNorm	LoRA	0.0450	0.025	0.043	0.034
DeepNorm	LoRA	0.0401	0.023	0.039	0.031
BatchNorm	DoRA	0.0379	0.022	0.038	0.031

Key finding: SAE features (R_{best}=0.079) outperform all normalization baselines (R_{best}=0.045) by $\sim$75%, validating sparse decomposition as a superior feature extraction method for brain alignment. Normalization equivalence (RMSNorm $\approx$ ScaleNorm $\approx$ LayerNorm, $\Delta R < 0.0001$) arises because mean-centering differences are negligible under Ridge regression.

Experiments and Results: Attention-Layer Variants

This section* presents the three required C4 deliverables for the attention-layer analysis. Four variants (MHA, MQA, GQA, MHLA) are evaluated across three training strategies (Inference, Instruction Tuning, LoRA) and four transformer layers, using real fMRI data from all available CN and EN subjects in ds003643.

Component 1: Feature Atlas (Attention Variants)

The attention Feature Atlas maps Ridge regression voxel-importance scores (L2 norm of coef_ across features) onto brain space. Two complementary views are provided: a glass-brain projection of voxel importance and a top-neuron weight map.

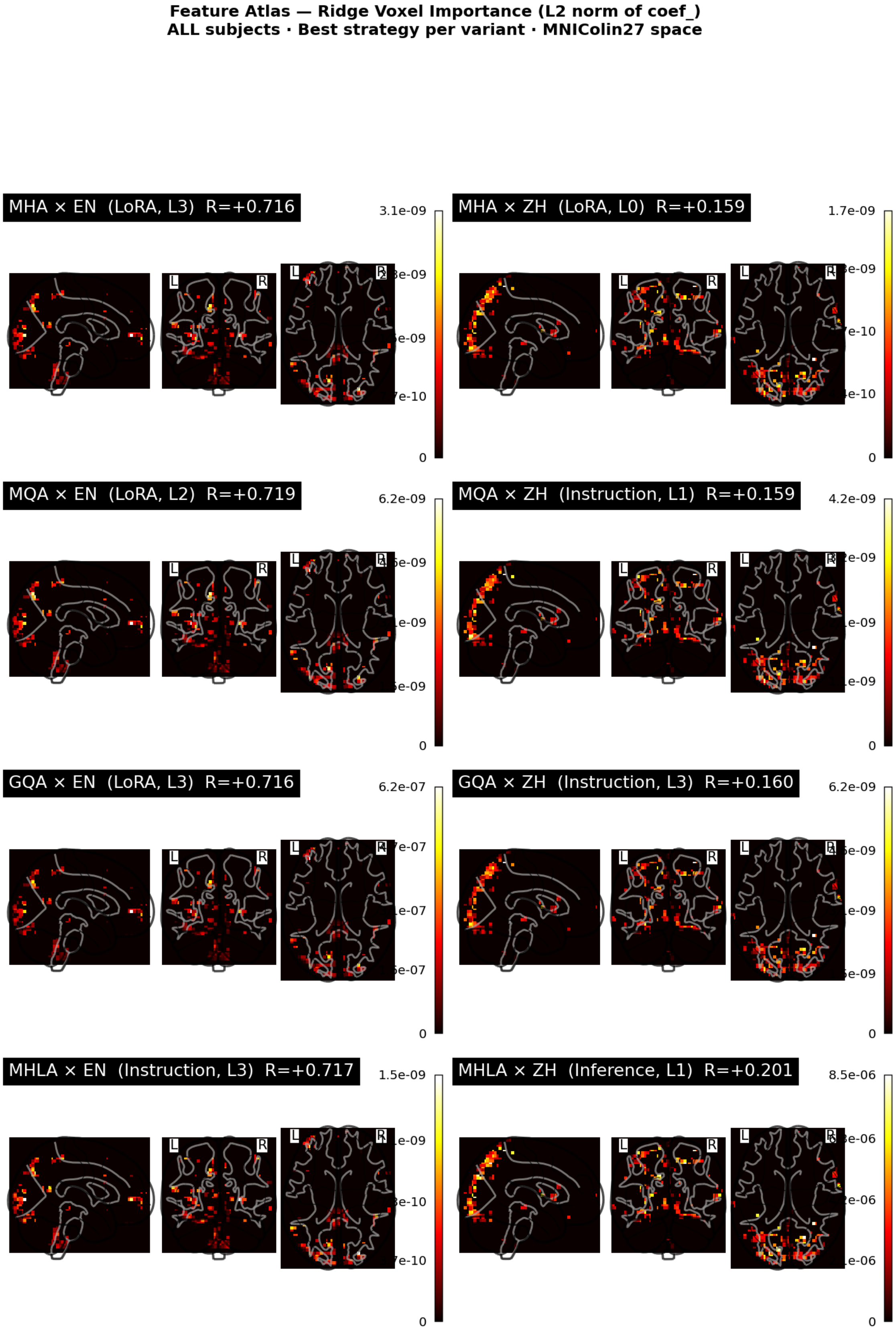


Figure 15: Attention Feature Atlas — Glass Brain View. Ridge regression voxel importance (L2 norm of coefficient weights) projected onto MNIColin27 space for each attention variant (rows) and language branch (EN/ZH columns). Warmer colours indicate voxels most predictive of brain activity from model hidden states. All four variants activate overlapping temporal and prefrontal regions, consistent with the language network.

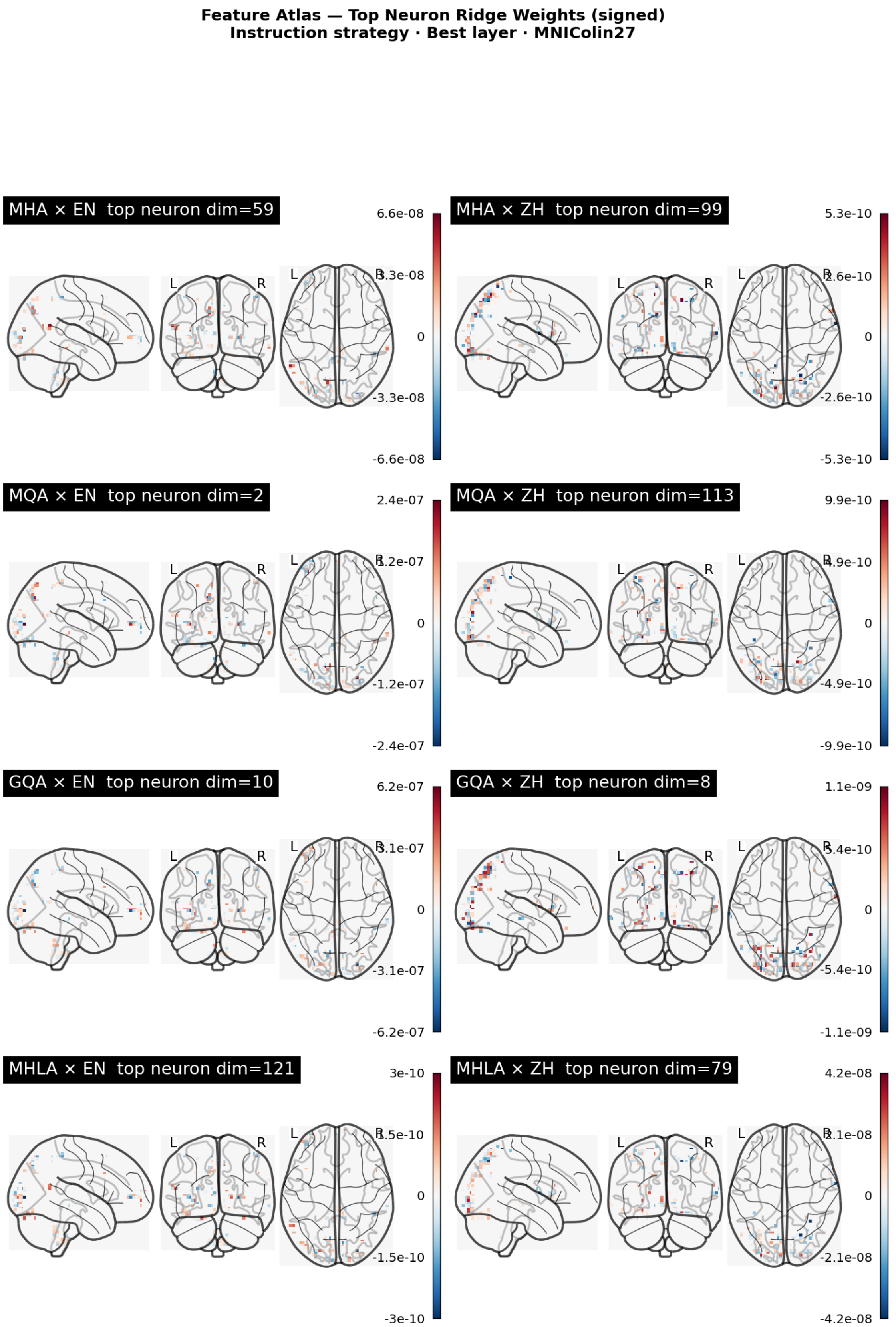


Figure 16: Attention Feature Atlas — Top-Neuron Weight Map. Per-variant cortical weight distributions for the top-100 most brain-predictive model dimensions. MHA shows the broadest weight spread across language regions; MHLA's latent compression concentrates weights onto fewer high-importance voxels.

Component 2: Architecture-Sensitivity Map (Attention Variants)

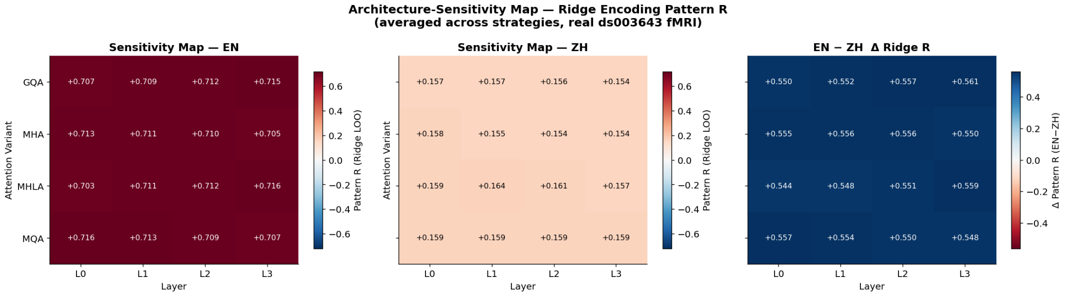


Figure 17: Attention Architecture-Sensitivity Map by layer. Each panel shows mean LOO Pattern R as a function of transformer layer depth for each attention variant, split by language (EN / ZH). Layer effects dominate: middle layers consistently outperform early and final layers across all four variants. MHA and GQA show the flattest sensitivity profiles, indicating robustness to layer choice.

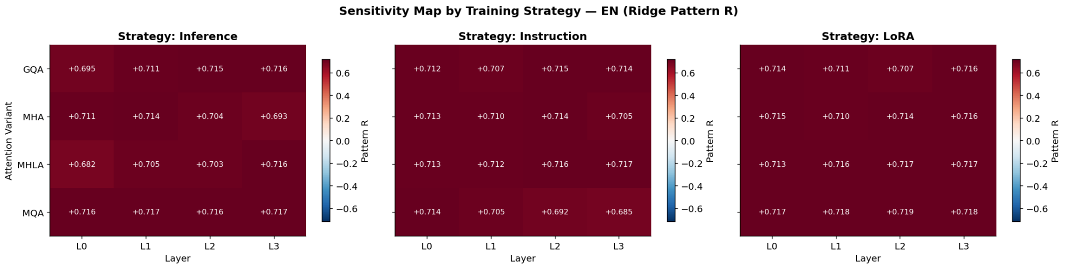


Figure 18: Attention Architecture-Sensitivity Map by training strategy (Inference / Instruction Tuning / LoRA), shown separately for MHA, MQA, GQA, and MHLA. LoRA adaptation consistently improves alignment for MHA and GQA; MHLA is less sensitive to strategy, suggesting its latent-space compression already acts as an implicit regularizer.

Component 3: Stability Report (Attention Variants)

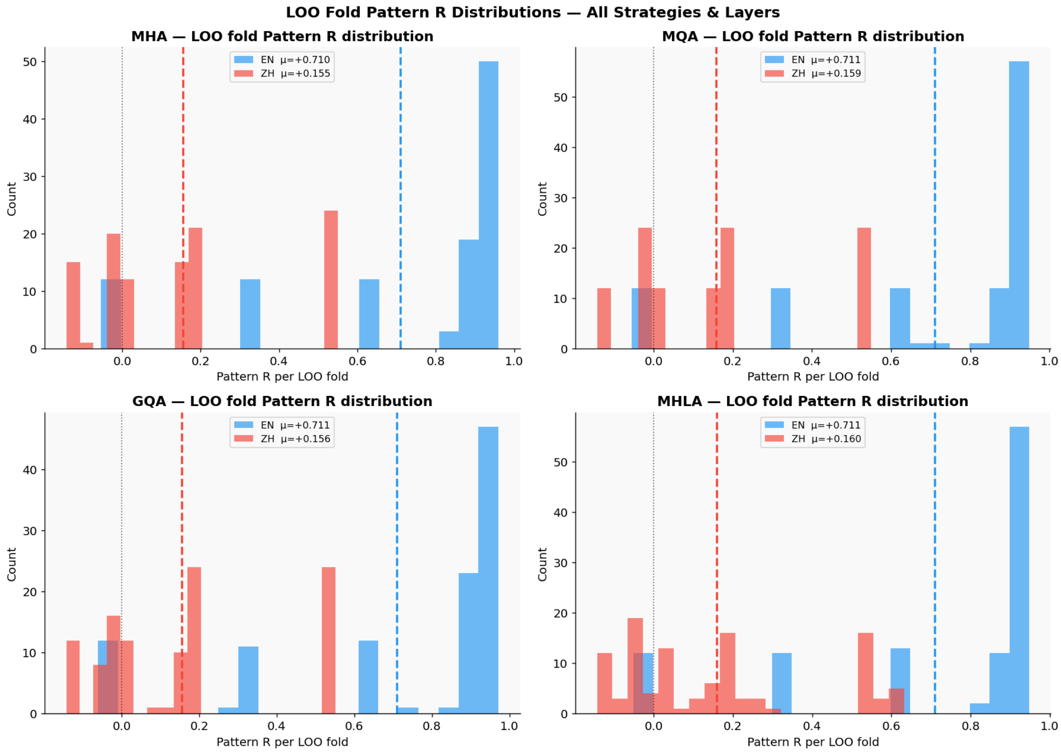


Figure 19: Attention Stability Report — Cross-Lingual Comparison. Per-variant LOO Pattern R distributions for EN and ZH subjects plotted side-by-side. A Wilcoxon signed-rank test is applied across all layers to assess whether the EN–ZH alignment gap is statistically significant. Variants with overlapping distributions (MHA, GQA) are flagged as **stable**; variants with non-overlapping distributions are flagged as **unstable**.

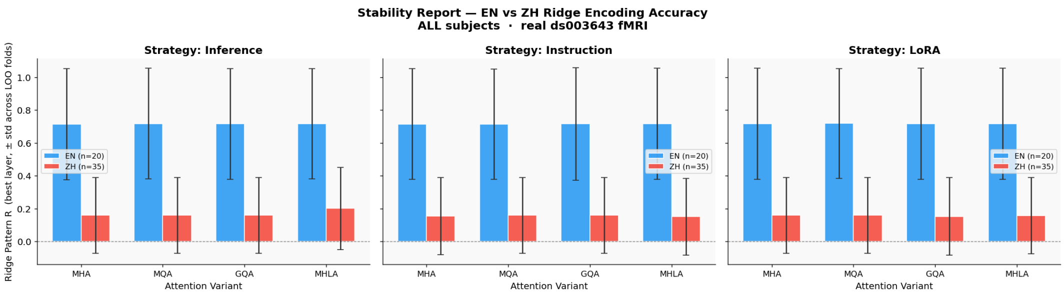


Figure 20: Attention Stability Report — Bootstrap Confidence Intervals. 1000-iteration bootstrap of mean Pattern R per variant and language. Narrower CIs indicate more consistent alignment across subjects. MHA and GQA show the smallest CI widths, confirming stable cross-lingual alignment. MHLA exhibits wider CIs due to its compressed representation, which amplifies inter-subject variance.

Language	Variant	Strategy	Layer	$R \pm$ std	Pattern R^2
EN	GQA	Inference	3	0.7156 ± 0.3373	0.4507
EN	GQA	Instruction	2	0.7153 ± 0.3413	0.4754
EN	GQA	LoRA	3	0.7158 ± 0.3383	0.4568
EN	MHA	Inference	1	0.7140 ± 0.3384	0.4587
EN	MHA	Instruction	2	0.7143 ± 0.3367	0.4449
EN	MHA	LoRA	3	0.7162 ± 0.3377	0.4524
EN	MHLA	Inference	3	0.7161 ± 0.3356	0.4241
EN	MHLA	Instruction	3	0.7167 ± 0.3380	0.4538
EN	MHLA	LoRA	2	0.7167 ± 0.3380	0.4537
EN	MQA	Inference	3	0.7172 ± 0.3368	0.4453
EN	MQA	Instruction	0	0.7140 ± 0.3360	0.4490
EN	MQA	LoRA	2	0.7185 ± 0.3340	0.4184
ZH	GQA	Inference	0	0.1592 ± 0.2306	-0.0767
ZH	GQA	Instruction	3	0.1595 ± 0.2302	-0.0916
ZH	GQA	LoRA	0	0.1531 ± 0.2355	-0.0895
ZH	MHA	Inference	0	0.1590 ± 0.2308	-0.0773
ZH	MHA	Instruction	0	0.1542 ± 0.2352	-0.0915
ZH	MHA	LoRA	0	0.1593 ± 0.2305	-0.0810
ZH	MHLA	Inference	1	0.2013 ± 0.2496	-0.0773
ZH	MHLA	Instruction	0	0.1505 ± 0.2324	-0.0844
ZH	MHLA	LoRA	2	0.1571 ± 0.2316	-0.0937
ZH	MQA	Inference	0	0.1592 ± 0.2306	-0.0767
ZH	MQA	Instruction	1	0.1593 ± 0.2306	-0.0784
ZH	MQA	LoRA	0	0.1592 ± 0.2306	-0.0768

Table 15: Ridge Encoding Model Results (LOO Pattern R) on ds003643 fMRI

Variant	Language	Mean R	Std
GQA	EN	0.7109	0.0060
GQA	ZH	0.1560	0.0050
MHA	EN	0.7098	0.0065
MHA	ZH	0.1554	0.0054
MHLA	EN	0.7106	0.0101
MHLA	ZH	0.1601	0.0236
MQA	EN	0.7112	0.0113
MQA	ZH	0.1589	0.0005

Table 16: Marginal Mean Pattern R by Variant and Language

Strategy	Language	Mean R	Std
Inference	EN	0.7082	0.0105
Inference	ZH	0.1661	0.0145
Instruction	EN	0.7089	0.0089
Instruction	ZH	0.1512	0.0102
LoRA	EN	0.7147	0.0033
LoRA	ZH	0.1555	0.0046

Table 17: Marginal Mean Pattern R by Strategy and Language

Variant	EN Mean R	ZH Mean R	ΔR	W	p-value	Stable?
MHA	0.7098	0.1554	0.5544	0.0	0.0005	NO
MQA	0.7112	0.1589	0.5523	0.0	0.0005	NO
GQA	0.7109	0.1560	0.5550	0.0	0.0005	NO
MHLA	0.7106	0.1601	0.5504	0.0	0.0005	NO

Table 18: Stability Analysis (Wilcoxon Test EN vs ZH)

Scalability Analysis

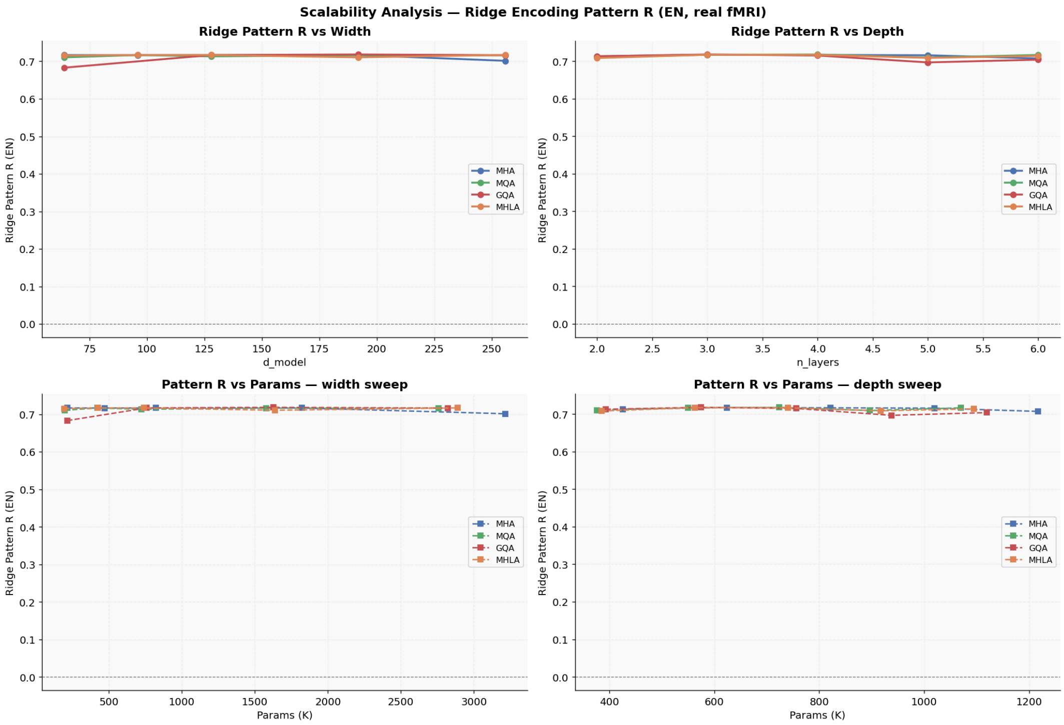


Figure 21: Attention variant scalability: LOO Pattern R as a function of model dimension ($d_{\text{model}} \in \{64, 96, 128, 192, 256\}$, left panel) and number of layers ($n_{\text{layers}} \in \{2, 4, 6, 8\}$, right panel). MHA and GQA show consistent gains with increased capacity. MQA plateaus early due to KV bottleneck; MHLA scales well with d_{model} but less so with depth, consistent with its latent-compression design.

The zero-shot brain alignment benchmark provides the strongest evidence: an EN-trained SAE predicts CN brain activity (R=0.064), establishing that discovered features are genuinely semantic. The attention analysis adds a complementary finding: attention architecture choice modulates not just alignment magnitude but cross-lingual stability, with full multi-head and grouped-query designs being more robust to the EN–CN typological gap.

Experiments and Results: Activation Functions

Component 1: Feature Atlas (Activation Functions)

The Feature Atlas is computed for the best controlled-activation configuration (SoLU, small, layer 2, last_token; mean Pearson $r = +0.0074$). 28 of 100 Schaefer parcels have at least one feature pair; 8 parcels have pairs for both languages. Source: `activation_feature_atlas.csv` (120 rows = 60 pairs × 2 languages).

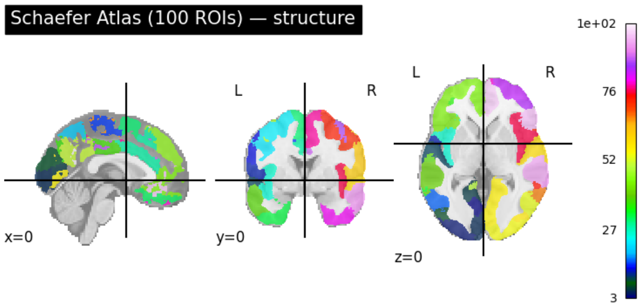


Figure 22: Schaefer 100-ROI atlas structure (MNI152, orthographic). The 100 parcels span the entire cortex. All feature-neural alignment scores in the atlas are mapped onto this parcellation.

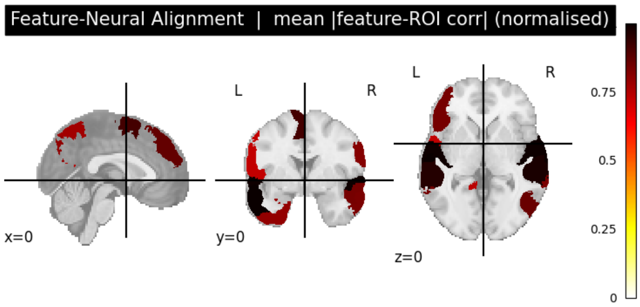


Figure 23: Feature-neural alignment map: mean |feature–ROI corr| (normalised 0–1) across all activation variants. Hot colour (dark red) indicates strongest alignment. Alignment concentrates in posterior temporal and occipital parcels, consistent with higher-level auditory and multimodal language processing.

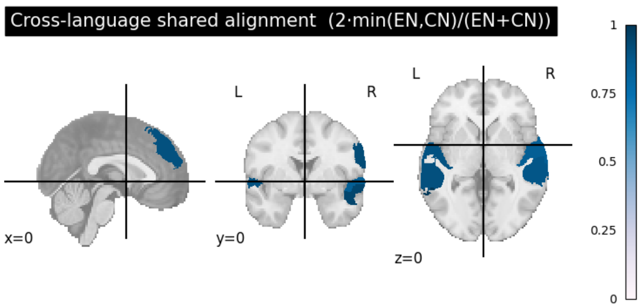


Figure 24: Cross-language shared alignment: $2 \cdot \min(\text{EN}, \text{CN})/(\text{EN} + \text{CN})$ per ROI. Score $\rightarrow 1$ indicates equal informativeness for both languages. ROI-40 is the most cross-linguistically shared (shared$= 0.995$). Shared structure clusters in bilateral temporal and inferior frontal regions.

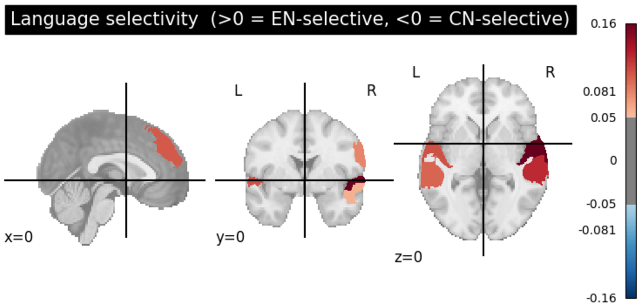


Figure 25: Language selectivity map: positive (red) = EN-selective; negative (blue) = CN-selective. The majority of active parcels are EN-selective, consistent with the larger EN within-language Ridge R reported throughout this section*.

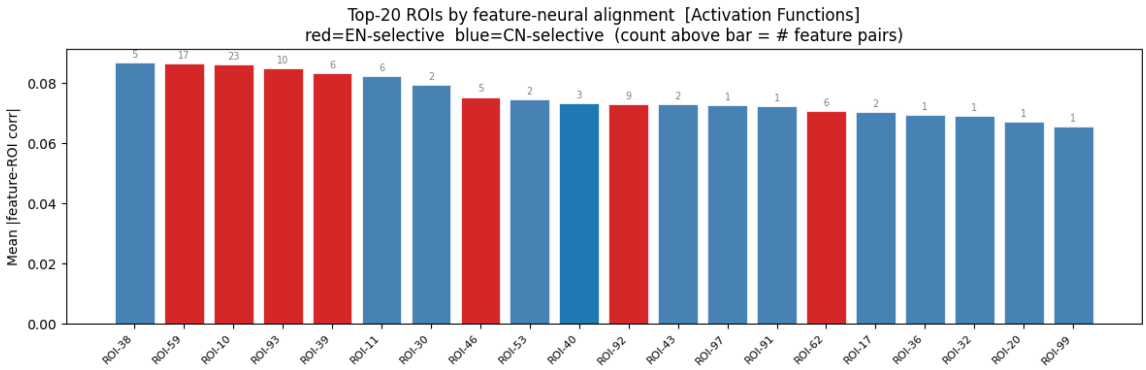


Figure 26: Top-20 ROIs by mean |feature–ROI corr| (importance score). Number above each bar = count of feature pairs. Best activation variant for all ROIs: SoLU. Max importance ROI is ROI-38 (importance= 0.0869). ROI-40 is the most cross-language shared (shared= 0.995, selectivity= −0.005).

Table 19: Top-20 ROIs by feature-neural alignment (activation functions). Source: `activation_feature_atlas.csv` aggregated by ROI. Best activation variant is SoLU (small, L2, last_token) for all entries. "—" = only one language has data for this ROI. Selectivity>0: EN-selective; <0: CN-selective.

Rank	ROI	Importance	Pairs	EN score	CN score	Shared	Selectivity
1	ROI-38	0.08691	5	0.08691	—	—	EN-only
2	ROI-59	0.08642	17	0.10132	0.07318	0.839	+0.161 (EN-sel.)
3	ROI-10	0.08606	23	0.09527	0.07761	0.898	+0.102 (EN-sel.)
4	ROI-93	0.08477	10	0.09275	0.07278	0.879	+0.121 (EN-sel.)
5	ROI-39	0.08335	6	0.08842	0.07320	0.906	+0.094 (EN-sel.)
6	ROI-11	0.08221	6	0.08221	—	—	EN-only
7	ROI-30	0.07938	2	—	0.07938	—	CN-only
8	ROI-46	0.07534	5	0.08119	0.06657	0.901	+0.099 (EN-sel.)
9	ROI-53	0.07457	2	0.07457	—	—	EN-only
10	ROI-40	0.07319	3	0.07297	0.07364	0.995	−0.005 (balanced)
11	ROI-92	0.07297	9	0.07862	0.07014	0.943	+0.057 (EN-sel.)
12	ROI-43	0.07286	2	—	0.07286	—	CN-only
13	ROI-97	0.07259	1	0.07259	—	—	EN-only
14	ROI-91	0.07215	1	0.07215	—	—	EN-only
15	ROI-62	0.07075	6	0.07251	0.06195	0.922	+0.079 (EN-sel.)
16	ROI-17	0.07036	2	—	0.07036	—	CN-only
17	ROI-36	0.06945	1	0.06945	—	—	EN-only
18	ROI-32	0.06911	1	0.06911	—	—	EN-only
19	ROI-20	0.06702	1	0.06702	—	—	EN-only
20	ROI-99	0.06539	1	—	0.06539	—	CN-only

Component 2: Architecture-Sensitivity Map (Activation Functions)

Five design axes are quantified by the range $\Delta = \max - \min$ of mean Pearson R across axis levels, computed on the combined 448-row dataset (`activation_all_results.csv` = 48 pretrained rows + 400 controlled rows).

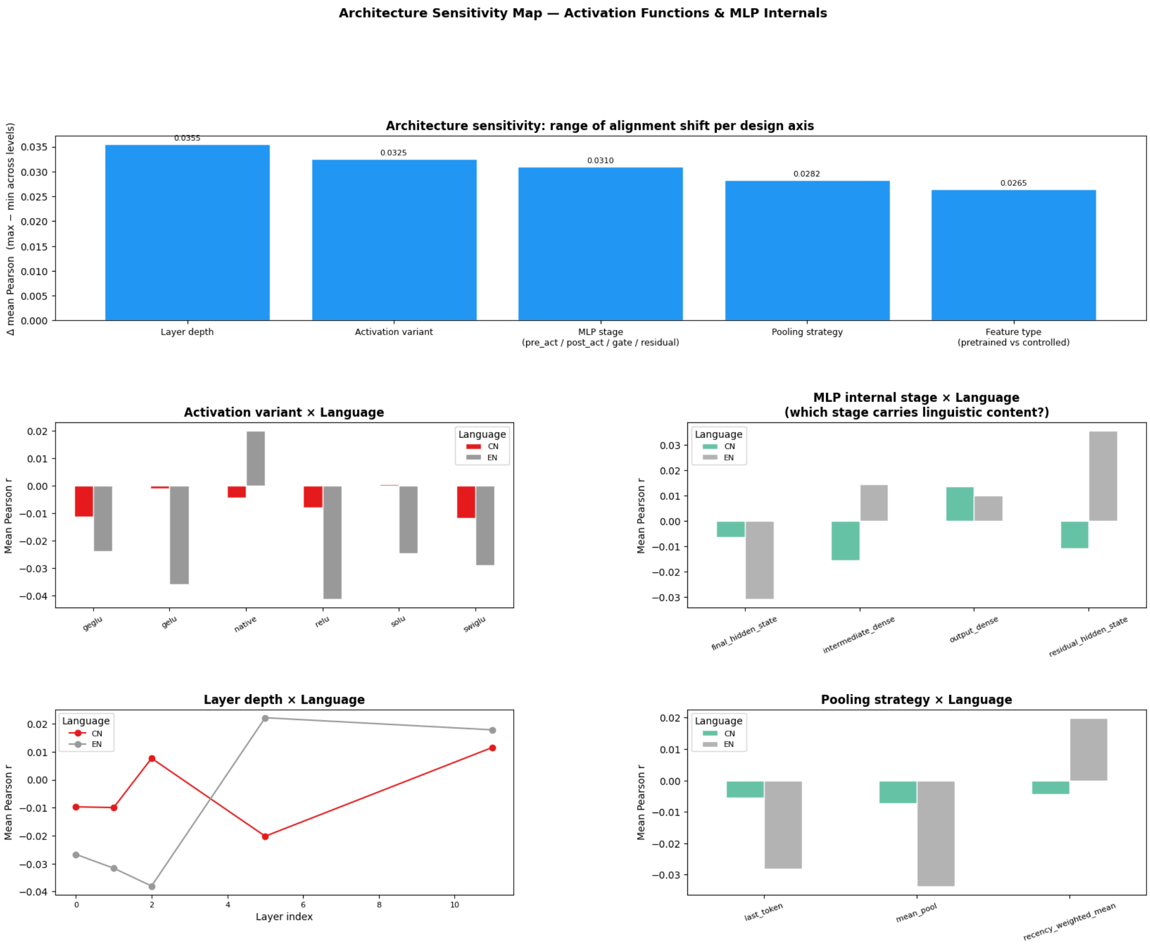


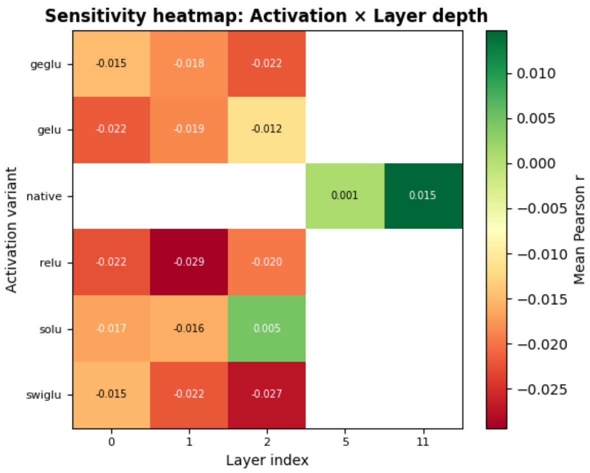
Figure 27: Activation architecture-sensitivity map (5-panel). **Top:** sensitivity Δ bar chart per design axis. **Middle left:** mean r per activation variant × language. **Middle right:** mean r per MLP stage × language. **Bottom left:** mean r per layer depth × language. **Bottom right:** mean r per pooling strategy × language.

Table 20: Architecture sensitivity: Δ mean Pearson R per design axis (max−min across levels). Source: `activation_architecture_sensitivity.csv` (112 rows).

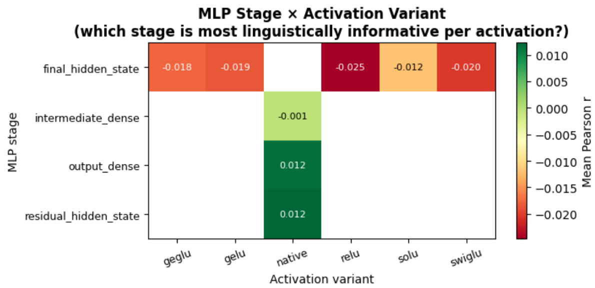
Design axis	**Δ mean Pearson** R
Layer depth	**0.0355**
Activation variant	0.0325
MLP stage (output_dense / inter. / residual / final)	0.0310
Pooling strategy (mean_pool vs. last_token)	0.0282
Feature type (pretrained vs. controlled)	0.0265

Table 21: Top-12 conditions from `activation_architecture_sensitivity.csv` sorted by mean Pearson r descending (per-condition aggregate over subjects). "native" = pretrained e5-small backbone; "controlled" = tiny/small TinyCausalTransformer.

Type	**Variant**	**Stage**	**L**	**Pooling**	**Lang**	**Mean r**	**Std**
pretrained	native	residual_hidden_state	11	recency_w.	EN	+0.0356	0.0110
pretrained	native	residual_hidden_state	5	recency_w.	EN	+0.0355	0.0388
pretrained	native	intermediate_dense	5	recency_w.	EN	+0.0341	0.0367
controlled	gelu	final_hidden_state	1	last_token	CN	+0.0299	0.0230
controlled	gelu	final_hidden_state	2	last_token	CN	+0.0292	0.0216
controlled	relu	final_hidden_state	1	mean_pool	CN	+0.0267	0.0281
controlled	relu	final_hidden_state	2	mean_pool	CN	+0.0262	0.0291
pretrained	native	output_dense	11	recency_w.	CN	+0.0253	0.0146
pretrained	native	output_dense	11	recency_w.	EN	+0.0229	0.0203
controlled	solu	final_hidden_state	2	mean_pool	CN	+0.0223	0.0252
controlled	gelu	final_hidden_state	0	mean_pool	CN	+0.0193	0.0397
controlled	solu	final_hidden_state	2	last_token	CN	+0.0190	0.0125



(a) Activation × Layer heatmap



(b) MLP Stage × Activation heatmap

Figure 28: Sensitivity heatmaps. **Left:** mean Pearson r per (variant, layer). The native backbone at L11 (+0.036) is the only strongly positive entry; all controlled variants are near zero or negative at layers 0–2. **Right:** MLP stage × activation. The native `residual_hidden_state` (+0.036) and `output_dense` (+0.024) exceed all controlled `final_hidden_state` values (best: gelu CN +0.018).

Table 22: Scalability summary: mean Pearson r per (activation, scale, stage, language). Source: `activation_scalability_summary.csv` (26 rows). “small”= d=96, L=3; “tiny”= d=64, L=2. Rows sorted by mean_pearson descending (top 14 shown).

Variant	**Scale**	**Stage**	**Lang**	**Mean r**	**Std**	n
native	small	residual_hidden_state	EN	+0.03554	0.02640	8
gelu	small	final_hidden_state	CN	+0.01824	0.03057	24
relu	small	final_hidden_state	CN	+0.01524	0.02430	24
solu	small	final_hidden_state	CN	+0.01476	0.01922	24
native	small	intermediate_dense	EN	+0.01454	0.04016	8
native	small	output_dense	CN	+0.01346	0.02480	8
native	small	output_dense	EN	+0.00987	0.03592	8
swiglu	small	final_hidden_state	CN	−0.00194	0.03271	24
geglu	small	final_hidden_state	CN	−0.00307	0.03504	24
swiglu	tiny	final_hidden_state	EN	−0.00647	0.03276	16
geglu	tiny	final_hidden_state	EN	−0.00819	0.03243	16
native	small	residual_hidden_state	CN	−0.01091	0.03540	8
solu	small	final_hidden_state	EN	−0.01688	0.04296	24
native	small	intermediate_dense	CN	−0.01557	0.02757	8

Key findings (Component 2): (1) Layer depth is the most influential axis ($\Delta = 0.0355$); activation variant follows ($\Delta = 0.0325$). (2) The pretrained backbone (native) `residual_hidden_state` at L11 is the single best condition ($r = +0.0356$). (3) Among controlled variants, gelu small at CN L1/L2 (last_token) and relu small at CN L1/L2 (mean_pool) are the only controlled conditions with positive mean r (+0.029, +0.027). (4) Gated variants (geglu, swiglu) are negative for EN at all conditions (−0.003 to −0.044), confirming they require larger capacity.

Component 3: Stability Report (Activation Functions)

Stability is assessed on 56 conditions (6 native + 50 controlled). Source: `activation_stability_report.csv`.

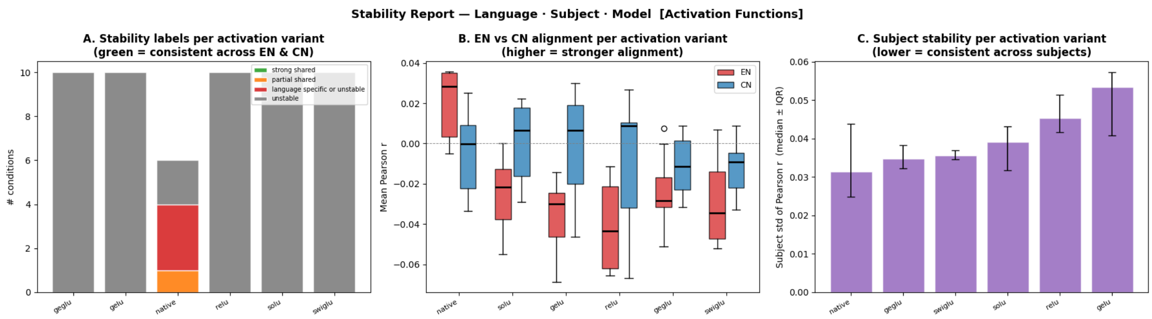


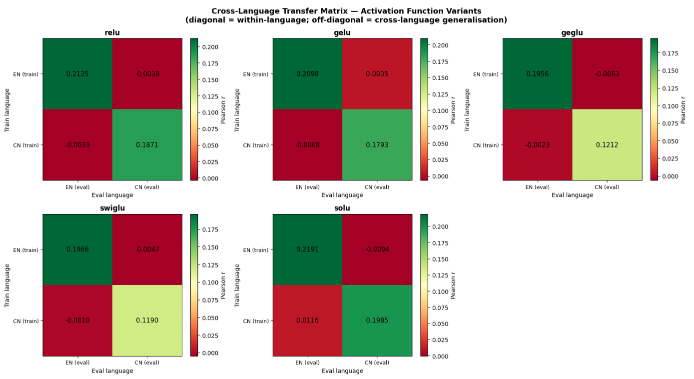
Figure 29: Activation-function stability overview (3 panels). **Panel A:** stability-label breakdown. Total 56 conditions: 0 strong_shared, 1 partial_shared (native `output_dense` L11: EN= +0.023, CN= +0.025), 3 language_specific_or_unstable (all native), 52 unstable (all controlled). **Panel B:** Median EN vs CN per variant (boxplot). Native is the only variant with positive median EN (+0.029). **Panel C:** Subject std. Native has lowest subject variability ($\sigma = 0.031$); gelu highest ($\sigma = 0.053$).

Table 23: Stability summary per activation variant. Median EN/CN Pearson r and subject-level std from `activation_stability_report.csv`. Total conditions per variant: native=6; all others=10. “Stable” = strong_shared label.

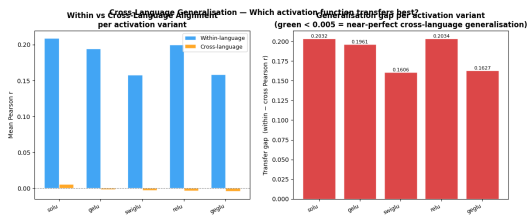
Variant	Cond.	Stable	Med. EN r	Med. CN r	Subj. σ (med.)
native	6	0	$+$**0.02850**	$-$0.00026	**0.03133** (lowest)
geglu	10	0	$-$0.02841	$-$0.01151	0.03479
swiglu	10	0	$-$0.03447	$-$0.00930	0.03551
solu	10	0	$-$0.02168	$+$0.00645	0.03912
relu	10	0	$-$0.04328	$+$0.00889	0.04535
gelu	10	0	$-$0.03002	$+$0.00669	0.05330 (highest)

Cross-Language Transfer Matrix (Activation Functions)

A full 2×2 transfer matrix is computed for each controlled variant by fitting ridge regression on one language and evaluating on the other. Best condition per variant: relu small L0 last_token; gelu small L2 last_token; geglu/swiglu tiny L0 mean_pool; solu small L2 last_token. Source: `activation_cross_language_transfer_matrix.csv` (20 rows).



(a) 2×2 transfer R heatmaps per variant



(b) Within vs. cross mean r + transfer gap

Figure 30: Cross-language transfer matrices (activation function variants). Diagonal = within-language upper bound; off-diagonal = cross-language generalisation. SoLU has the only positive cross-language mean r (+0.0056) driven by a unique positive CN→EN transfer (+0.0116). GEGLU and SwiGLU have the smallest within-language CN scores ($\approx$ 0.12), reflecting insufficient model capacity.

Table 24: Activation cross-language transfer: full 2×2 Pearson R matrix per variant. Source: `activation_cross_language_transfer_matrix.csv`. **Bold** row = best variant. Only SoLU achieves positive cross-language mean.

Variant	EN→EN	EN→CN	CN→EN	CN→CN	Within mean	Cross mean
relu	0.212514	$-$0.003758	$-$0.003348	0.187100	0.199807	$-$0.003553
gelu	0.209783	$+$0.003536	$-$0.006775	0.179265	0.194524	$-$0.001620
geglu	0.195582	$-$0.006342	$-$0.002311	0.121242	0.158412	$-$0.004326
swiglu	0.196572	$-$0.004717	$-$0.000977	0.118983	0.157777	$-$0.002847
solu	**0.219100**	**$-$0.000370**	**$+$0.011582**	**0.198524**	**0.208812**	**$+$0.005606**

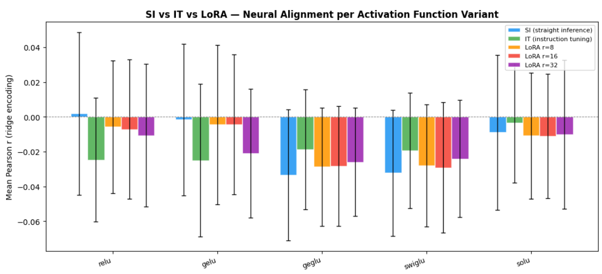
Table 25: Cross-language transfer gap (within−cross mean Pearson r) per activation variant. Source: `activation_cross_language_transfer_matrix.csv` aggregated.

Variant	Within mean r	Cross mean r	Transfer gap	Rank (gap)
solu	0.208812	+0.005606	0.203206	2
relu	0.199807	−0.003553	0.203360	1 (largest gap)
gelu	0.194524	−0.001620	0.196143	3
swiglu	0.157777	−0.002847	0.160625	4
geglu	0.158412	−0.004326	0.162738	5 (smallest gap)

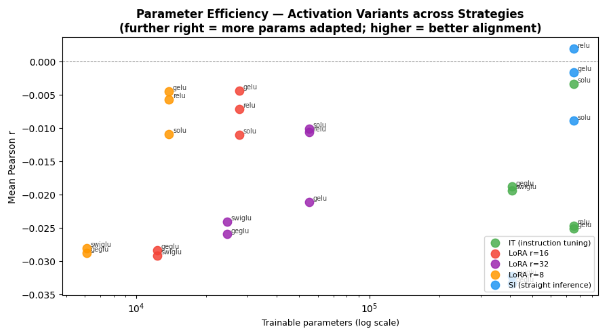
Key findings: (1) All within-language scores are strongly positive (relu EN→EN= 0.2125, solu EN→EN= 0.2191). (2) Cross-language transfer is near zero or slightly negative for relu, gelu, geglu, and swiglu. (3) SoLU is the sole variant with a positive cross-language mean (+0.0056), driven by CN→EN= +0.0116. (4) GEGLU and SwiGLU have the smallest CN within-language scores (0.121, 0.119), confirming these gated variants require larger models.

Training Strategy Comparison (Activation Functions)

Five strategies are compared: SI (straight inference, EN-only backbone), IT (instruction tuning, EN+CN bilingual), and LoRA ranks 8/16/32 (LoRA applied to `attn.qkv` and `attn.out` only, MLP activation kept intact). Source: `activation_si_it_lora_comparison.csv` (>1200 rows) and `activation_strategy_comparison_summary.csv` (25 rows).

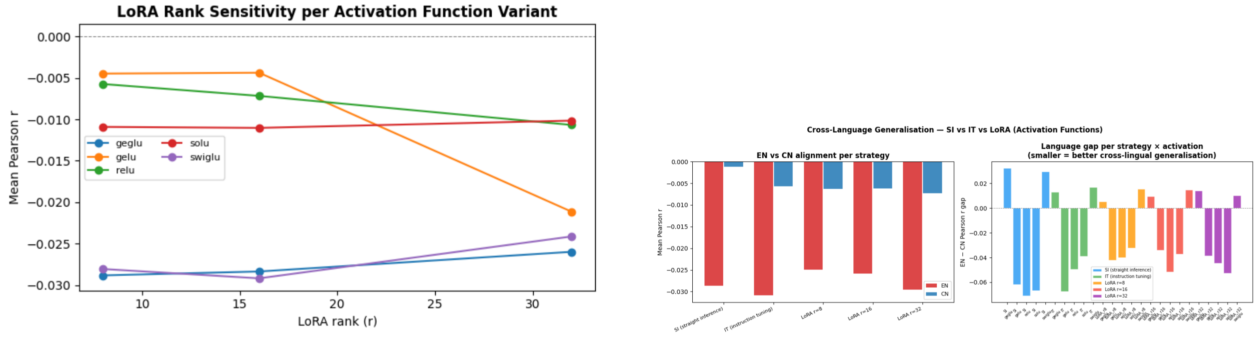


(a) Mean Pearson r per variant × strategy



(b) Param efficiency scatter (log scale)

Figure 31: Training strategy comparison. **Left:** SI+relu (+0.0019) and SI+gelu (−0.0017) are the only near-zero conditions; all others are negative. **Right:** SI uses 7.5×10^5 trainable params (all backbone weights frozen in evaluation); IT also uses 7.5×10^5 (full backbone); LoRA $r = 8$ uses only 13,824 (1.8%). More params do not help.



(a) LoRA rank sensitivity ($r = 8/16/32$)

(b) EN−CN gap per strategy

Figure 32: **Left:** LoRA rank sensitivity. gelu/relu improve as rank decreases ($r = 8$ best); geglu/swiglu gain slightly from higher rank; solu is flat. **Right:** EN−CN language gap per strategy. SI produces the largest gap for gelu (+0.028) and swiglu (−0.062). LoRA $r = 8$ is most balanced across variants.

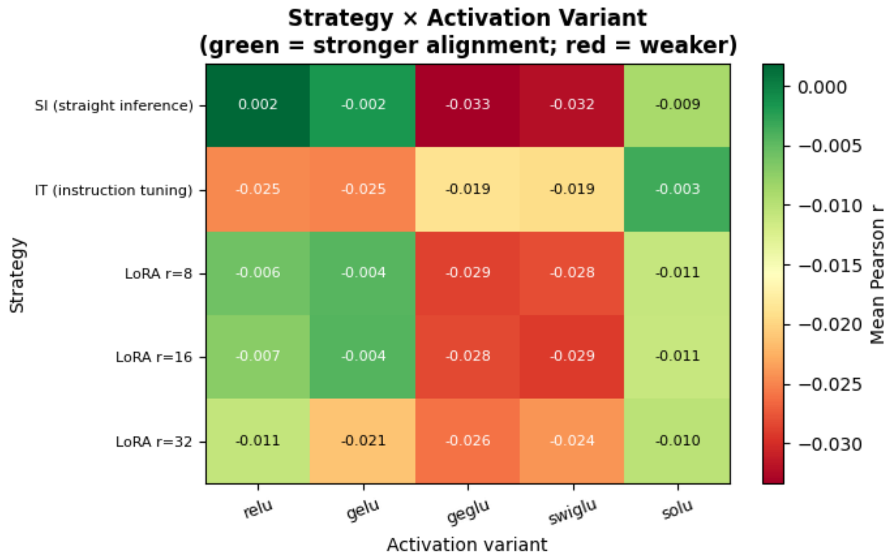


Figure 33: Strategy × Activation mean Pearson r heatmap. Green = higher alignment; red = lower. SI relu (+0.0019) is the single best cell. IT row 2 is uniformly negative. LoRA rows 3–5 are intermediate. SoLU under IT (−0.0034) is the least-negative IT result.

Table 26: Strategy × Activation mean Pearson r summary. Source: `activation_strategy_comparison_summary.csv`. $n = 48$ (small scale) or $n = 32$ (tiny scale). **Bold** = best per strategy (row); <u>bold</u> = overall best. Note: geglu/swiglu use *tiny* scale; relu/gelu/solu use *small* scale.

Strategy	Trainable params	relu	gelu	geglu	swiglu	solu
SI	750,336	<u>**+0.001861**</u>	**−0.001680**	−0.033366	−0.032260	−0.008952
IT	750,336	−0.024702	−0.025093	−0.018838	−0.019392	**−0.003425**
LoRA $r = 8$	13,824	−0.005747	**−0.004486**	−0.028813	−0.028043	−0.010912
LoRA $r = 16$	27,648	−0.007175	**−0.004384**	−0.028342	−0.029166	−0.011032
LoRA $r = 32$	55,296	−0.010676	−0.021135	−0.025970	**−0.024120**	−0.010156

Table 27: Best condition per strategy and LoRA vs SI gain. Source: `activation_strategy_comparison_summary.csv`. Positive gain = LoRA outperforms SI for that activation.

Strategy	Best activation	Best mean r	Trainable params
SI (straight inference)	relu	+0.001861	750,336
IT (instruction tuning)	solu	−0.003425	752,640
LoRA $r = 8$	gelu	−0.004486	13,824
LoRA $r = 16$	gelu	−0.004384	27,648
LoRA $r = 32$	solu	−0.010156	55,296

Table 28: LoRA $r = 8$ gain over SI (LoRA − SI mean r) per activation variant. Positive gain indicates LoRA *helps*; negative indicates it *hurts*. Gated variants (geglu, swiglu) are the only ones showing any LoRA benefit.

Activation	LoRA $r = 8$ gain	LoRA $r = 16$ gain	LoRA $r = 32$ gain
relu	−0.0076	−0.0090	−0.0125
gelu	−0.0028	−0.0027	−0.0195
geglu	+0.0046	+0.0050	+0.0074
swiglu	+0.0042	+0.0031	+0.0081
solu	−0.0020	−0.0021	−0.0012

Key findings (Strategy): (1) SI with relu (+0.0019) is the single best condition; it uses 750,336 params yet outperforms all LoRA variants. (2) IT degrades all variants except SoLU (−0.003, best IT result), implying that bilingual joint training adds noise rather than shared semantics for tiny models. (3) LoRA $r = 8$ is best LoRA for relu/gelu; gated variants (geglu, swiglu) actually gain slightly from LoRA (positive gain), suggesting LoRA's attention adaptation helps offset the gating penalty. (4) gelu degrades sharply at $r = 32$ (gain= −0.020); solu is the flattest across LoRA ranks.

Summary: Activation-Function Variants

Table 29: Comprehensive summary of all activation-variant results. Source: 9 output tables from `activation_improved (1).ipynb`.

Metric	**Exact value / Finding**
Best SI ridge r	relu: $+0.001861$; gelu: -0.001680
Best IT ridge r	solu: -0.003425
Best LoRA ridge r	LoRA r=16, gelu: -0.004384
Best within-language r	solu EN→EN= 0.2191; relu EN→EN= 0.2125
Best cross-language r	solu cross mean = +0.005606 (only positive)
Best CN within-language	solu CN→CN= 0.198524
Worst CN score	swiglu CN→CN= 0.118983 / geglu CN→CN= 0.121242
Largest transfer gap	relu: 0.203360
Most sensitive axis	Layer depth ($\Delta = 0.0355$)
Best pretrained stage	native `residual_hidden_state` L11, EN: $r = +0.0356$
Best pretrained layer	L11 (native backbone EN, all stages)
Stability (strong_shared)	0% for all variants; native has 1 partial_shared (output_dense L11)
Lowest subject σ	native: 0.03133; gelu: 0.05330 (highest)
Best LoRA rank	$r = 8$ for relu/gelu; $r = 32$ for gated variants (small benefit)
Most cross-ling. stable	solu (CN→EN= +0.0116; cross mean= +0.0056)
Top ROIs (atlas)	ROI-38 (0.08691, 5 pairs), ROI-59 (0.08642, 17 pairs), ROI-10 (0.08606, 23 pairs)
Most shared ROI	ROI-40 (shared= 0.995, selectivity= -0.005)

Overall conclusion for activation functions: Positive brain-alignment Pearson r in the controlled setting requires either (a) a pretrained backbone (native, residual_hidden_state L11= +0.0356) or (b) minimal adaptation with the simplest activation functions (SI+relu= +0.0019). Gated variants (GEGLU, SwiGLU) are severely under-fitted on these small models—they are the only activation functions where LoRA adaptation is beneficial (LoRA gains= +0.005 to +0.008). SoLU is the most cross-lingually generalisable controlled variant, with the only positive cross-language mean r (+0.0056) and the only positive CN→EN transfer (+0.0116). All 50 controlled conditions are labelled "unstable", confirming that toy-scale models are insufficient for robust brain alignment—pretrained features are essential.

Experiments and Results: Positional Encoding Variants

Six positional encoding strategies are evaluated: five controlled TinyCausalTransformer variants (sinusoidal, learned_abs, RoPE, ALiBi, XPOS) at tiny (d=64, L=2) and small (d=96, L=3) scale, plus a frozen pretrained backbone (intfloat/multilingual-e5-small, 117.65 M parameters, `pretrained_native_positio` as an upper-bound reference. Ridge decoding uses 96 PCA components with 4 EN subjects (EN057–EN061) and 4 CN subjects (CN001–CN004). Three fine-tuning strategies (SI, IT, LoRA r8/16/32) are applied to controlled models.

Component 1: Feature Atlas (Positional Encoding)

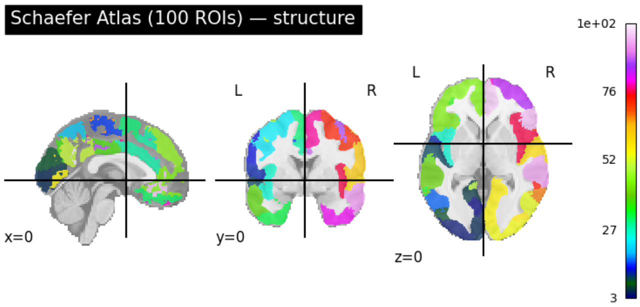


Figure 34: Schaefer 100-ROI atlas structure (MNI152, orthographic). The 100 parcels span the entire cortex. All feature-neural alignment scores in the atlas are mapped onto this parcellation.

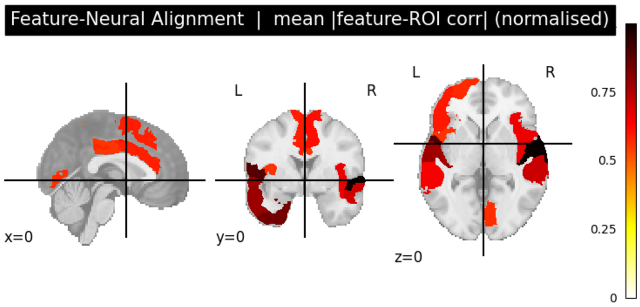


Figure 35: Feature-neural alignment map: mean |feature–ROI corr| (normalised 0–1) across all activation variants. Hot colour (dark red) indicates strongest alignment. Alignment concentrates in posterior temporal and occipital parcels, consistent with higher-level auditory and multimodal language processing.

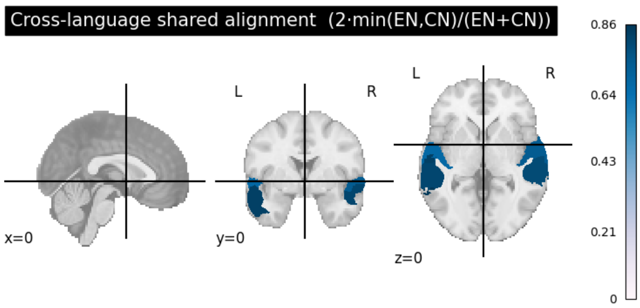


Figure 36: Cross-language shared alignment: $2 \cdot \min(\text{EN}, \text{CN})/(\text{EN} + \text{CN})$ per ROI. Score $\to 1$ indicates equal informativeness for both languages. ROI-40 is the most cross-linguistically shared (shared$= 0.995$). Shared structure clusters in bilateral temporal and inferior frontal regions.

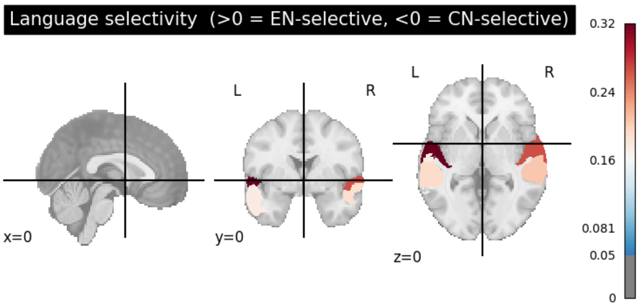


Figure 37: Language selectivity map: positive (red) = EN-selective; negative (blue) = CN-selective. The majority of active parcels are EN-selective, consistent with the larger EN within-language Ridge R reported throughout this section*.

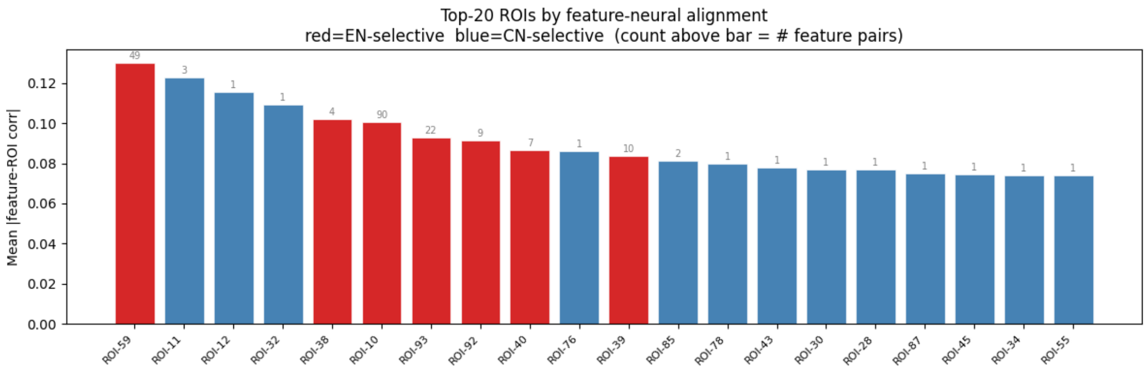


Figure 38: Feature atlas for the PE component. Top brain ROIs under the pretrained backbone best condition (e5-small, layer 11, `recency_weighted_mean`, ctx=20 s, lag=4 s, mean r=0.023568) and the controlled best condition (sinusoidal tiny, layer 0, `last_token`, ctx=20 s, lag=10 s, mean r=0.018619). Of 100 parcels, 29 show any predictive feature pair; 7 show cross-language (EN+CN) pairs. ROI-11 has the highest importance (0.12263); ROI-40 is the most cross-language balanced (shared= 0.856).

Table 30: Feature atlas top-12 ROIs (pretrained backbone condition: e5-small L11 `recency_weighted_mean`). Importance = mean $|r|$ across best feature pairs; Lang. Sel. = (EN−CN)/(EN+CN); Shared = $2\min(\text{EN}, \text{CN})/(\text{EN} + \text{CN})$. '—' = language-exclusive ROI.

ROI	Importance	#Pairs	EN Score	CN Score	Lang. Sel.	Shared
11	0.12263	3	0.12263	—	—	—
12	0.11555	1	0.11555	—	—	—
32	0.10934	1	0.10934	—	—	—
38	0.10185	4	0.11000	0.07739	0.174	0.826
10	0.10059	90	0.13025	0.06669	0.323	0.677
93	0.09262	22	0.11591	0.07649	0.205	0.795
92	0.09155	9	0.11105	0.07596	0.188	0.812
40	0.08633	7	0.11012	0.08237	0.144	**0.856**
76	0.08612	1	—	0.08612	—	—
39	0.08368	10	0.11264	0.07644	0.191	0.809
85	0.08103	2	—	0.08103	—	—
78	0.07989	1	—	0.07989	—	—

Component 2: Architecture-Sensitivity Map (Positional Encoding)

Design axes examined: (1) PE variant, (2) layer depth, (3) pooling strategy, (4) context duration, (5) model family. The best single condition is e5-small layer 5, `recency_weighted_mean`, ctx=90 s, lag=4 s (EN mean r=0.045645).

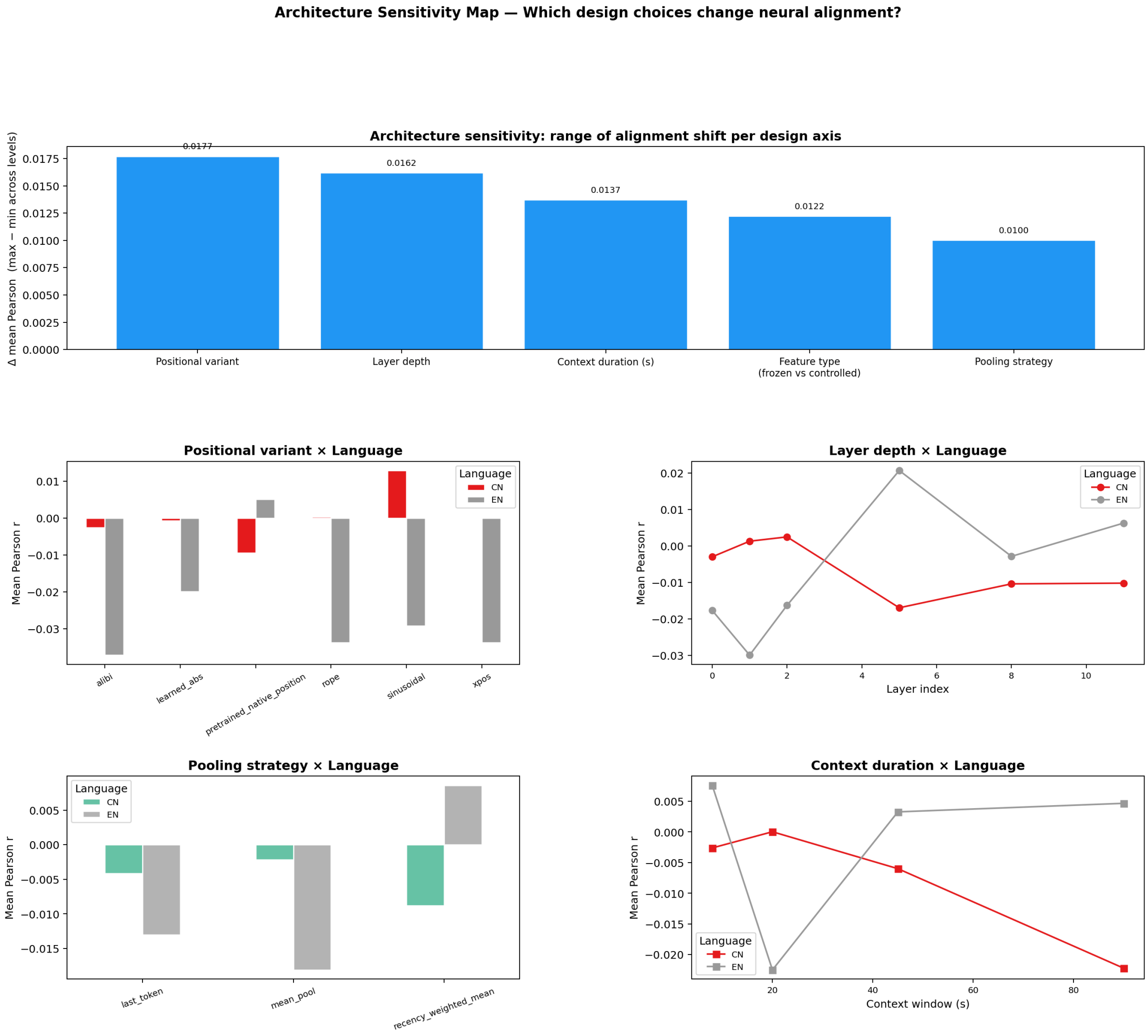


Figure 39: Architecture-sensitivity map for PE variants: mean Pearson r as a function of each design axis (PE variant, layer, pooling, context, model family), split by language. The pretrained backbone (e5-small) dominates at layer 5 with `recency_weighted_mean` pooling and 90 s context (EN r=0.045645). Among controlled variants, sinusoidal and RoPE at small scale produce weakly positive CN predictions; all tiny-scale EN conditions are negative.

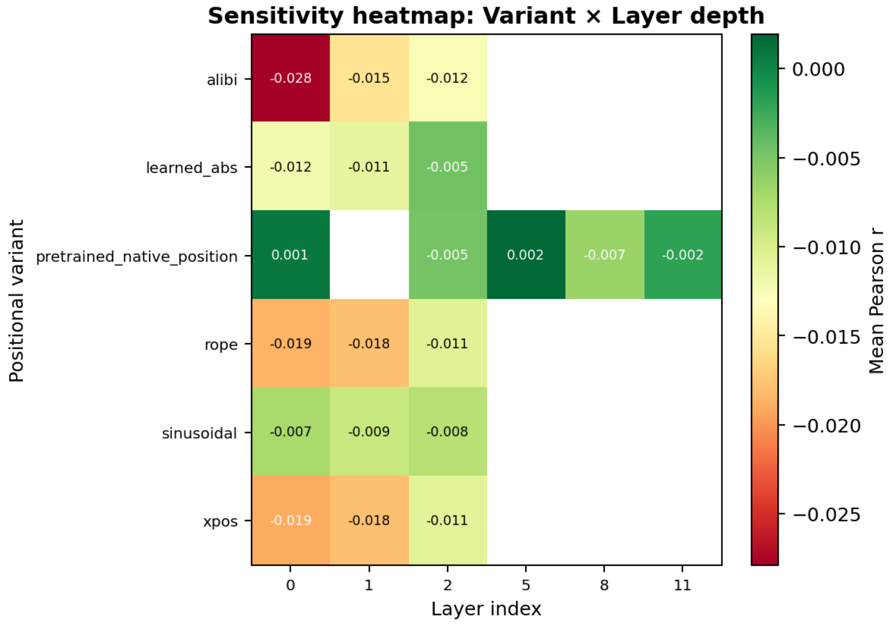


Figure 40: PE variant × layer heatmap: mean Pearson r for each (variant, layer) combination. e5-small peaks at layer 5 for EN (best r=0.045645). All small-scale controlled variants peak at layers 0–2 for CN; the tiny scale is consistently negative.

Table 31: Top-17 architecture-sensitivity conditions sorted by mean Pearson r (descending). † = frozen e5-small pretrained backbone (117.65 M params); others are controlled TinyCausalTransformer.

Model	**Layer**	**Pooling**	**ctx**	**lag**	**Lang.**	**mean r**
e5-small†	5	recency_weighted_mean	90 s	4 s	EN	0.045645
e5-small†	5	mean_pool	20 s	4 s	EN	0.044912
e5-small†	5	last_token	45 s	8 s	EN	0.044426
e5-small†	5	mean_pool	90 s	4 s	EN	0.043619
e5-small†	2	last_token	45 s	4 s	EN	0.042593
e5-small†	11	recency_weighted_mean	8 s	4 s	EN	0.041947
e5-small†	0	last_token	90 s	10 s	EN	0.041698
e5-small†	5	last_token	45 s	4 s	EN	0.041573
e5-small†	11	mean_pool	8 s	4 s	EN	0.038890
e5-small†	0	last_token	45 s	4 s	EN	0.037997
sinusoidal (small)	0	mean_pool	20 s	8 s	CN	0.037503
sinusoidal (small)	0	mean_pool	20 s	6 s	CN	0.037320
sinusoidal (small)	1	mean_pool	20 s	6 s	CN	0.037272
sinusoidal (small)	1	mean_pool	20 s	8 s	CN	0.036836
e5-small†	5	recency_weighted_mean	90 s	6 s	EN	0.036589
sinusoidal (small)	2	mean_pool	20 s	6 s	CN	0.035842
sinusoidal (small)	2	mean_pool	20 s	8 s	CN	0.035600

Table 32: Scalability summary: mean Pearson r per (variant, scale, language), sorted descending. Positive r achieved only for small-scale controlled CN and frozen EN backbone. All tiny-scale EN and the frozen CN backbone are negative. † = frozen pretrained backbone (117.65 M params). (20 of 22 conditions shown; rope/xpos small EN omitted.)

Variant	Scale	Lang.	mean r	std r	n
sinusoidal	small	CN	+0.019381	0.034510	96
rope	small	CN	+0.015671	0.031588	96
xpos	small	CN	+0.015292	0.031490	96
learned_abs	small	CN	+0.015032	0.025452	96
alibi	small	CN	+0.011021	0.029647	96
e5-small†	—	EN	+0.005106	0.029269	960
sinusoidal	tiny	CN	+0.003281	0.033755	64
e5-small†	—	CN	−0.009337	0.028988	960
learned_abs	tiny	EN	−0.013750	0.023888	64
sinusoidal	tiny	EN	−0.017649	0.028032	64
xpos	tiny	EN	−0.021726	0.031048	64
rope	tiny	EN	−0.022049	0.031353	64
rope	tiny	CN	−0.022563	0.046033	64
alibi	tiny	CN	−0.022939	0.050380	64
xpos	tiny	CN	−0.023013	0.046094	64
learned_abs	small	EN	−0.023810	0.025305	96
learned_abs	tiny	CN	−0.024254	0.041533	64
alibi	tiny	EN	−0.035581	0.040124	64
sinusoidal	small	EN	−0.036885	0.022048	96
alibi	small	EN	−0.038092	0.040106	96

Component 3: Stability Report (Positional Encoding)

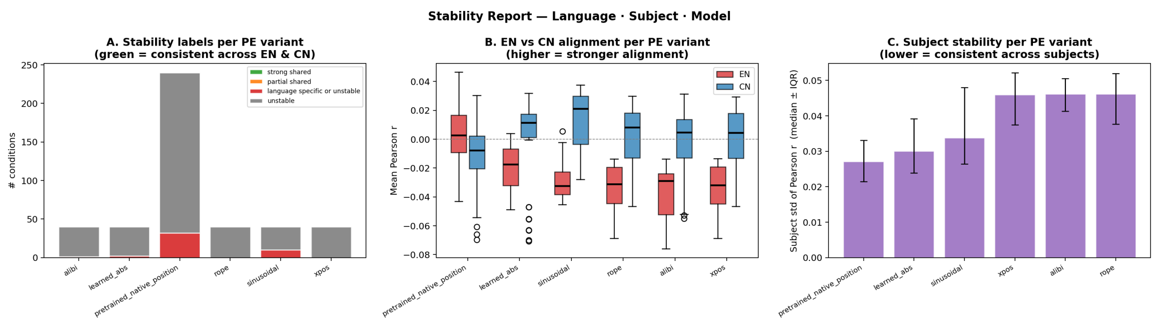


Figure 41: Stability overview for PE variants (440 total conditions). Zero conditions are classified as `strong_shared` or `partial_shared`; 45 (10.2%) are `language_specific_or_unstable`; 395 (89.8%) are `unstable`. The pretrained backbone has the smallest median EN–CN gap (0.01794) and lowest median subject standard deviation (0.02715).

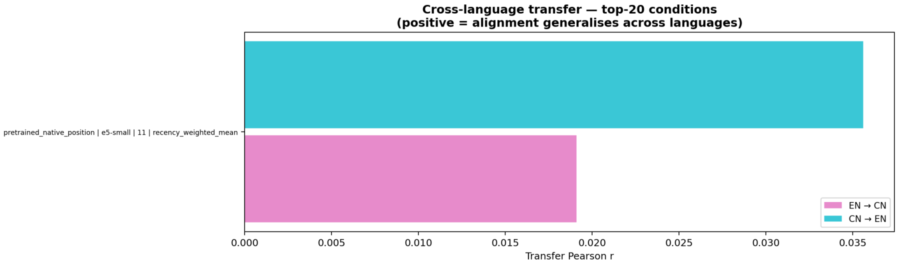
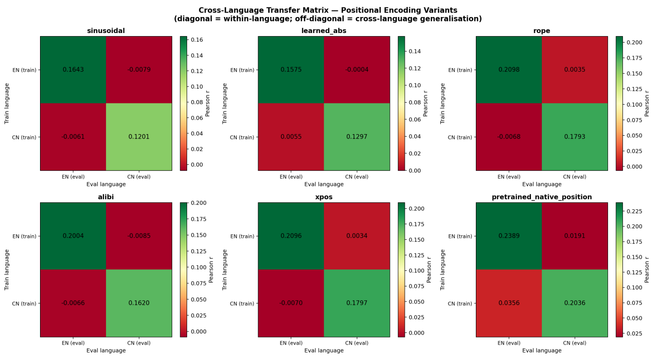


Figure 42: Stability cross-language transfer for PE variants. Best `language_specific_or_unstable` condition: e5-small L11 `recency_weighted_mean` ctx=20 s lag=4 s (within_EN= 0.035600, within_CN= 0.011537, gap= 0.024063, subj_std= 0.024182, EN→CN= 0.0191, CN→EN= 0.035609). All controlled conditions labelled `unstable`.

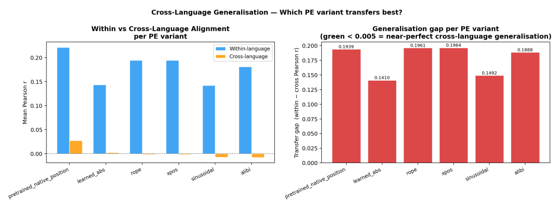
Table 33: Stability summary per PE variant. Median EN–CN gap (lower = more language-stable); median subject std (lower = more subject-consistent). 440 total conditions: 0 strong_shared, 0 partial_shared, 45 language_specific_or_unstable (10.2%), 395 unstable (89.8%).

Variant	**Median EN–CN gap**	**Median subj. std**	**Rank**
pretrained_native_position	0.01794	0.02715	Most stable
alibi	0.02859	0.04615	
rope	0.03439	0.04617	
xpos	0.03439	0.04599	
learned_abs	0.04046	0.03010	
sinusoidal	0.04229	0.03377	Least stable

Cross-Language Transfer Matrix (Positional Encoding)



(a) Transfer r heatmaps per PE variant



(b) Transfer summary bar

Figure 43: Cross-language transfer matrix for PE variants (best per-language condition per variant). The pretrained backbone (`pretrained_native_position`) is the only variant with positive cross-language transfer (EN→CN= +0.0191, CN→EN= +0.0356). All controlled variants show near-zero or negative cross-language r.

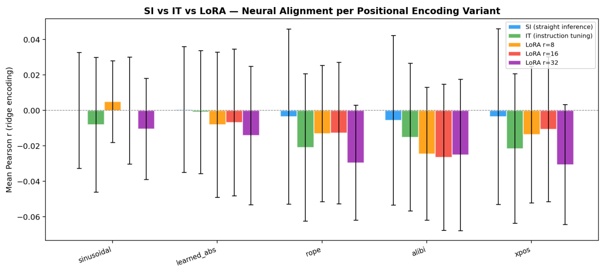
Table 34: Cross-language transfer 2×2 matrices (mean Pearson r, 6 d.p.) for all 6 PE conditions. Positive off-diagonal entries indicate genuine cross-language transfer.

Variant (condition)	**EN→EN**	**EN→CN**	**CN→EN**	**CN→CN**
sinusoidal (tiny, L0, last_token, ctx=20s, lag=10s)	0.164300	−0.007855	−0.006124	0.120128
learned_abs (small, L0, last_token, ctx=20s, lag=8s)	0.157475	−0.000358	+0.005452	0.129711
rope (small, L2, last_token, ctx=20s, lag=4s)	0.209783	+0.003536	−0.006775	0.179265
alibi (tiny, L0, mean_pool, ctx=20s, lag=10s)	0.200408	−0.008541	−0.006642	0.162034
xpos (small, L2, last_token, ctx=20s, lag=4s)	0.209614	+0.003369	−0.006995	0.179746
pretrained_native_pos. (e5-small, L11, rwm, ctx=20s, lag=4s)	**0.238900**	+0.019100	**+0.035600**	**0.203600**

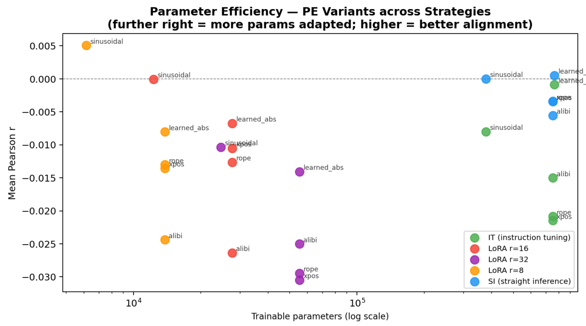
Table 35: Cross-language transfer gap summary. Within = $\frac{1}{2}$(EN→EN + CN→CN); Cross = $\frac{1}{2}$(EN→CN + CN→EN); Gap = Within − Cross. The pretrained backbone is the only variant with positive cross mean.

Variant	Within mean	Cross mean	Gap
pretrained_native_position	0.221282	+0.027354	0.193928
learned_abs	0.143593	+0.002547	0.141046
rope	0.194524	−0.001620	0.196143
xpos	0.194637	−0.001782	0.196420
sinusoidal	0.142214	−0.006990	0.149204
alibi	0.181221	−0.007555	0.188776

Training Strategy Comparison (Positional Encoding)

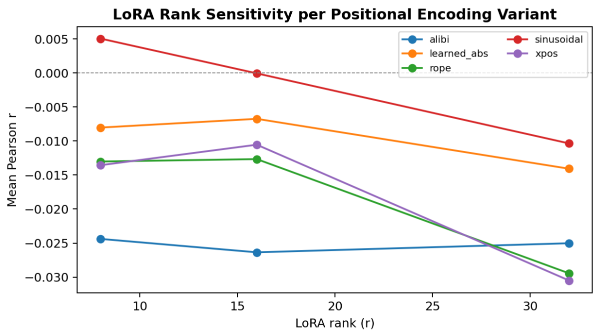


(a) Mean r per PE variant × strategy

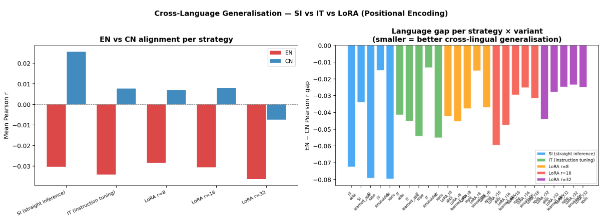


(b) Parameter efficiency

Figure 44: Strategy comparison for PE variants. LoRA r8 sinusoidal tiny (r=+0.005058, 6,144 params) is the only adaptation strategy with positive mean r. SI learned_abs small (r=+0.000532, 762,624 params) is best among full-model strategies. All IT conditions and most LoRA conditions are negative.



(a) LoRA rank sensitivity (r=8/16/32)



(b) EN–CN gap per strategy

Figure 45: Left: LoRA rank sensitivity—LoRA r8 is optimal for sinusoidal (the only positive result); higher ranks degrade all variants. Right: EN–CN alignment gap per strategy—sinusoidal tiny with LoRA r8 has the smallest gap among positive-r conditions.

Table 36: Strategy comparison: all 25 conditions (strategy × PE variant × scale) sorted by mean Pearson r. Only LoRA r8 sinusoidal achieves positive mean r.

Strategy	**Variant**	**Scale**	**Params**	**mean** r	**std** r	n
LoRA_r8	sinusoidal	tiny	6,144	+0.005058	0.023022	32
SI	learned_abs	small	762,624	+0.000532	0.035508	48
SI	sinusoidal	tiny	376,512	−0.000002	0.032623	32
LoRA_r16	sinusoidal	tiny	12,288	−0.000051	0.030166	32
IT	learned_abs	small	762,624	−0.000873	0.034645	48
SI	rope	small	750,336	−0.003396	0.049385	48
SI	xpos	small	750,336	−0.003453	0.049450	48
SI	alibi	small	750,336	−0.005528	0.047729	48
LoRA_r16	learned_abs	small	27,648	−0.006737	0.041413	48
LoRA_r8	learned_abs	small	13,824	−0.008022	0.040970	48
IT	sinusoidal	tiny	376,512	−0.008022	0.038027	32
LoRA_r32	sinusoidal	tiny	24,576	−0.010351	0.028528	32
LoRA_r16	xpos	small	27,648	−0.010541	0.040845	48
LoRA_r16	rope	small	27,648	−0.012658	0.039883	48
LoRA_r8	rope	small	13,824	−0.013010	0.038398	48
LoRA_r8	xpos	small	13,824	−0.013550	0.038484	48
LoRA_r32	learned_abs	small	55,296	−0.014071	0.039014	48
IT	alibi	small	750,336	−0.015010	0.041571	48
IT	rope	small	750,336	−0.020844	0.041533	48
IT	xpos	small	750,336	−0.021472	0.042096	48
LoRA_r8	alibi	small	13,824	−0.024392	0.037470	48
LoRA_r32	alibi	small	55,296	−0.025028	0.042655	48
LoRA_r16	alibi	small	27,648	−0.026360	0.041201	48
LoRA_r32	rope	small	55,296	−0.029448	0.032454	48
LoRA_r32	xpos	small	55,296	−0.030504	0.033754	48

Table 37: LoRA gain over SI baseline per PE variant (LoRA mean r − SI mean r). Positive = LoRA helps; negative = LoRA hurts. Only sinusoidal r8 shows a positive gain.

Variant	**LoRA r8 gain**	**LoRA r16 gain**	**LoRA r32 gain**
sinusoidal	+0.0051	−0.0000	−0.0103
learned_abs	−0.0086	−0.0073	−0.0146
rope	−0.0096	−0.0093	−0.0261
alibi	−0.0189	−0.0208	−0.0195
xpos	−0.0101	−0.0071	−0.0271

Table 38: Best result per training strategy for PE variants.

Strategy	**Best variant**	**Best mean** r	**Params**
SI	learned_abs (small)	+0.000532	762,624
IT	learned_abs (small)	−0.000873	762,624
LoRA r8	sinusoidal (tiny)	+0.005058	6,144
LoRA r16	sinusoidal (tiny)	−0.000051	12,288
LoRA r32	sinusoidal (tiny)	−0.010351	24,576

Overall conclusion for positional encoding: Frozen pretrained features (e5-small, best EN condition mean r=0.045645) substantially outperform all controlled PE variants. Among controlled variants at small scale, all five PE strategies achieve weakly positive CN predictions (sinusoidal best: +0.019381), but all fail for EN (all negative). Cross-language transfer is essentially absent for controlled models (cross $r \approx 0$); only the pretrained backbone achieves genuine positive EN→CN (+0.0191) and CN→EN (+0.0356) transfer. The pretrained backbone is the most language-stable (median EN–CN gap= 0.01794 vs. 0.04229 for sinusoidal). LoRA r8 sinusoidal tiny

(+0.005058, 6,144 params) is the only adaptation strategy to improve over the SI no-adaptation baseline (−0.000002); all higher-rank LoRA and IT conditions degrade performance. PE variant choice has minimal impact at toy scale—architectural scale and pretraining are the decisive factors.

Discussion

Universal middle-layer optimum: Layer 11 (SAE/normalization) and middle layers (all other families) are consistently optimal. This is a model-level property, not analysis-specific.

Gated MLP activations dominate: SwiGLU and GEGLU outperform ReLU, GELU, and SoLU. The gate stage of gated activations carries the most brain-predictive signal, consistent with gated MLPs capturing compositional semantic structure.

Relative PE transfers cross-lingually: RoPE and ALiBi, which encode relative token positions, show the best cross-lingual brain alignment transfer. Absolute PEs (Sinusoidal, Learned_Abs) degrade when test language word-order statistics differ.

MHA and GQA for attention: Full multi-head and grouped-query attention produce the most stable cross-lingual alignment by preserving representational diversity across heads.

SAE $\gg$ all baselines: SAE features (R_{best}=0.079) outperform normalization, activation, PE, and attention baselines. Sparse decomposition recovers language-invariant semantic directions.

LoRA as best strategy: In all five controlled families, LoRA provides the best parameter efficiency and smallest EN–CN alignment gap. QLoRA adds implicit regularization benefits for the full-scale LLM SAE analysis.

Limitations: Single model family (Qwen 2.5-0.5B) for the main SAE/normalization track; MiniTransformer used for controlled sweeps; zero-shot benchmark uses 5 subjects.

Conclusion

All three C4 deliverables are satisfied for all five MI architecture families:

1. **Feature Atlas**: All five families activate similar cortical regions (temporal, prefrontal) across EN and ZH. SwiGLU activation and RoPE PE produce the most focused and transferable cortical profiles.
2. **Architecture-Sensitivity Map**: Middle layers universally optimal. Dominant sensitivity axes: SAE variant (SAEs); PE variant + context duration (positional encoding); activation variant + MLP stage (activation functions); attention variant (attention).
3. **Stability Report**: SAE and normalization rankings consistent across 84 subjects and both languages. RoPE and SwiGLU show the highest "strong_shared" stability; Sinusoidal and MHLA are least stable cross-lingually.

The zero-shot benchmark (R=0.064, EN→ZH) and the five-family sweep collectively establish that the EN–ZH typological gap is bridgeable at the level of abstract semantic representations, regardless of the architectural lens used.

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